



SCIENTIFIC COMMITTEE
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Assessment workflow demo: ALB

WCPFC-SCXX-XXXX/SA-WP-XX (WORKFLOW DEMO)
XXX

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Table of contents

1	Executive summary	2
2	Introduction	2
3	Data compilation	2
4	Model description	3
5	Model development	3
6	Results	3
7	Reporting examples	4
8	Discussion	4
9	Tables	4
10	Figures	8
11	Supplementary outputs	51
12	Appendix: supplementary methods	52
13	References	52

1 Executive summary

This is a workflow demonstration, not a stock assessment. It uses ALB outputs to demonstrate the model-to-report tools. The diagnostic-model figures use the archived MFCL-compatible KIMSA fit from job 24781. Stepwise compares completed native MFCL runs (INI 1003 and INI 1007) with the same KIMSA baseline. Starting-value, self-test, retrospective, sensitivity, ensemble, profile, projection, hindcast and additional-output figures use illustrative display values to demonstrate report construction. They are not new assessment estimates.

Table 1 links the report and all interactive supplements. Replace the illustrative study bundles with fitted results, write the assessment interpretation in the section files, and update the report cover before submission.

2 Introduction

3 Data compilation

The diagnostic model retains the ALB observation timing and two-region structure. Catch, abundance-index, length-composition and conditional age-at-length observations are available in the interactive diagnostic supplement.

Region boundaries follow the [2024 South Pacific albacore assessment, Rev.03](#), Section 4.1 and Figure 2. The spatial comparison also shows its Figure 1 reproduction of the 2021 structure.

Table 2 describes the fishery roles and observation periods. Figure 6 shows data availability. Index observations and predictions follow the model likelihood's geometric-mean centering convention.

4 Model description

The archived ALB fit uses the MFCL-compatible age-region state, age-based selectivity and five-pass hybrid F, without tagging. Its original F upper bound of 3 is retained for reproduction; new model specifications default to 6.

Length compositions use `square_fit` with fishery divisors and a raw-record sample-size cap of 1000.

Figure 8, Figure 9 and Figure 10 describe the biological inputs and estimates; Figure 33 shows fishery selectivity. Additional-format panels illustrate output types that are not estimated by this ALB reference model.

5 Model development

Figure 11 compares the latest completed matching ALB baselines found in Kflow: native MFCL 2.2.7.0 / INI 1003 (job 24514), native MFCL 2.2.7.9 / INI 1007 (job 24515), then KIMSA / hybrid F (job 24781). All used TunaFlow 2.5. The [interactive stepwise comparison](#) shows population quantities and model fits together by selected step. These population panels use the archived annual report censuses; they are distinct from the time-weighted annual summaries of the detailed diagnostic snapshot.

6 Results

Figure 32 shows annual reference-model trajectories. Stocks use time-weighted annual means; recruitment and catches use annual totals. Depletion is the ratio of annual spawning-potential summaries. Index points and predictions use annual geometric means. The interactive viewer opens annual summaries; select Original to inspect the fitted time steps.

Length-composition fits (Figure 14; Figure 17) show observed sample counts as bars and predictions as lines, pooled over years on the original full-bin support. Length-residual panels retain the fitted records and compressed-tail support. Conditional age-at-length fits are pooled by region using raw sample counts, preserving the model's prediction scale.

Index fits and residuals (Figure 13), length-composition residuals and conditional age-at-length fits (Figure 20) are generated from the same result bundle used by the interactive viewer.

The following studies demonstrate the intended diagnostic presentation: starting-value trials (Figure 21), simulation-estimation recovery (Figure 24), retrospective analysis (Figure 26), objective profiles (Figure 27) and sensitivity comparisons. Their displayed values are illustrative. Hindcasting (Figure 62; Table 13) uses annual observed indices, five annual rolling training cutoffs, and forecasts up to three years ahead. The demonstration fits log-linear index trends independently at each cutoff; exponentiated predictions are conditional medians. MASE uses only training naive errors for scaling. These are not refitted stock-assessment models.

7 Reporting examples

The ensemble and management-quantity panels demonstrate the reporting formats. Projection panels compare fixed-catch and F scenarios using a simple surplus-production kernel initialized at the ALB biomass scale; its dynamics and variability are illustrative. They do not establish ALB stock status. Supply actual reference points, model retention decisions, uncertainty draws and projection results before writing management conclusions.

8 Discussion

9 Tables

Table 1: Public report, interactive supplements and published outputs for this workflow demonstration.

Resource	Figures and tables	Interactive output
Workflow demonstration	PDF report	HTML report
Diagnostic fit	HTML supplement	Model viewer
Stepwise development	HTML supplement	Compare development steps
Jitter	HTML supplement	Model viewer
Self-test	HTML supplement	Model viewer
Retrospective analysis	HTML supplement	Model viewer
Sensitivities	HTML supplement	Model viewer
Uncertainty ensemble	HTML supplement	Model viewer
Objective profiles	HTML supplement	Model viewer
Additional output examples	HTML supplement	Model viewer
Hindcasting	HTML supplement	Model viewer
Projection scenarios	HTML supplement	Model viewer
Published outputs	Public repository	Rendered artifacts

Table 2: Fishery definitions and observation units.

Fishery	Region	Role	First year	Last year	Unit
1	1	Catch	1954	2022	fish
2	1	Catch	1954	2022	fish
3	1	Catch	1954	2022	fish
4	1	Catch	1954	2022	fish
5	1	Catch	1992	2022	fish
6	1	Catch	1982	2022	fish
7	1	Catch	1987	2022	fish
8	1	Catch	1954	2022	fish
9	1	Catch	1984	2022	fish

Fishery	Region	Role	First year	Last year	Unit
10	1	Catch	1985	2022	fish
11	1	Catch	1982	2022	t
12	1	Catch	1987	2022	t
13	1	Catch	1983	1990	t
14	2	Catch	1957	2022	fish
15	2	Catch	1961	2021	fish
16	2	Catch	1963	2021	fish
17	2	Catch	1986	2006	t
18	1	Index	1954	2022	relative index
19	1	Index	1992	2022	relative index
20	2	Index	1954	2022	relative index

Table 3: Illustrative output. Dirichlet–multinomial composition settings.

Fishery	Likelihood	Group	Theta
Fishery A	dirichlet_multinomial	1	20

Table 4: Illustrative output. Groups sharing composition-weight parameters.

Group	Fishery
1	Fishery A

Table 5: Illustrative output. Parameters on or near their declared bounds.

Parameter	Estimate	Lower	Upper
Steepness	0.99	0.2	0.99

Table 6: Illustrative output. Parameter correlations from the supplied covariance estimate.

Parameter	Partner	Correlation
log R0	Index q	-0.97
Initial abundance	Recruitment scale	0.96

Table 7: Management quantities and supplied uncertainty summaries.

Quantity	Period	Median	50% interval	80% interval	95% interval
$SB/SB_{F=0}$	2016–2020	0.45	0.41–0.49	0.37–0.53	0.33–0.57

Quantity	Period	Median	50% interval	80% interval	95% interval
F/F_{MSY}	2016–2020	0.68	0.61–0.75	0.54–0.82	0.46–0.90
SB/SB_{MSY}	2016–2020	1.35	1.22–1.48	1.08–1.62	0.94–1.76

Table 8: Spawning-depletion comparisons for the declared reference periods.

Period	Reference	Ratio
2016–2020	2012–2015	0.95

Table 9: Stock-status probabilities for the declared reference points.

Condition	Probability
$SB/SB_{F=0} < 0.20$	0.01
$F/F_{MSY} > 1$	0.05
$SB/SB_{MSY} < 1$	0.04

Table 10: Model reference points and their definitions.

Quantity	Median	Lower	Upper	Unit
MSY	60000	48000	72000	t/year
F_{MSY}	0.3	0.24	0.36	per year
SB_{MSY}	1000000	800000	1200000	t
SB_{MSY}/SB_0	0.28	0.22	0.34	ratio

Table 11: Comparison of supplied uncertainty summaries.

Source	Quantity	Median	Lower	Upper
Structural	$SB/SB_{F=0}$	0.45	0.39	0.51
Structural	F/F_{MSY}	0.68	0.57	0.79
Structural and estimation	$SB/SB_{F=0}$	0.45	0.35	0.55
Structural and estimation	F/F_{MSY}	0.68	0.51	0.85

Table 12: Illustrative projected quantities in the final projection year; medians and central 80% intervals.

Scenario	Quantity	Year	Median	Lower	Upper	Unit
fixed-catch	biomass	2052	0.9499	0.9473	0.952	ratio
	depletion					

Scenario	Quantity	Year	Median	Lower	Upper	Unit
fixed-catch	catch	2052	45150	45150	45150	t
fixed-catch	fishing	2052	0.01518	0.01514	0.01522	per year
	mortality					
fixed-catch	total	2052	2997000	2989000	3004000	t
	biomass					
higher-F	biomass	2052	0.7427	0.7378	0.7519	ratio
	depletion					
higher-F	catch	2052	180200	179000	182400	t
higher-F	fishing	2052	0.08	0.08	0.08	per year
	mortality					
higher-F	total	2052	2343000	2328000	2372000	t
	biomass					
lower-F	biomass	2052	0.8678	0.8609	0.8719	ratio
	depletion					
lower-F	catch	2052	107400	106500	107900	t
lower-F	fishing	2052	0.04	0.04	0.04	per year
	mortality					
lower-F	total	2052	2738000	2716000	2751000	t
	biomass					

Table 13: Held-out prediction bias (predicted minus observed), RMSE and MASE by series and forecast lead. N counts scored forecasts; MASE uses each cutoff’s training naive-error scale.

Series	Lead	N	Bias	RMSE	MASE	Unit
Index 18 (Region 1)	1	5	-0.08847	0.10778	0.39718	relative
Index 18 (Region 1)	2	4	-0.095363	0.117	0.42491	relative
Index 18 (Region 1)	3	3	-0.093805	0.12075	0.41723	relative
Index 19 (Region 1)	1	5	-0.0691	0.18227	0.52469	relative
Index 19 (Region 1)	2	4	-0.0245	0.13014	0.41413	relative
Index 19 (Region 1)	3	3	-0.026698	0.1508	0.49854	relative
Index 20 (Region 2)	1	5	-0.2195	0.25108	0.84494	relative
Index 20 (Region 2)	2	4	-0.26202	0.28574	0.99863	relative

Series	Lead	N	Bias	RMSE	MASE	Unit
Index 20 (Region 2)	3	3	-0.30353	0.32467	1.1481	relative

10 Figures

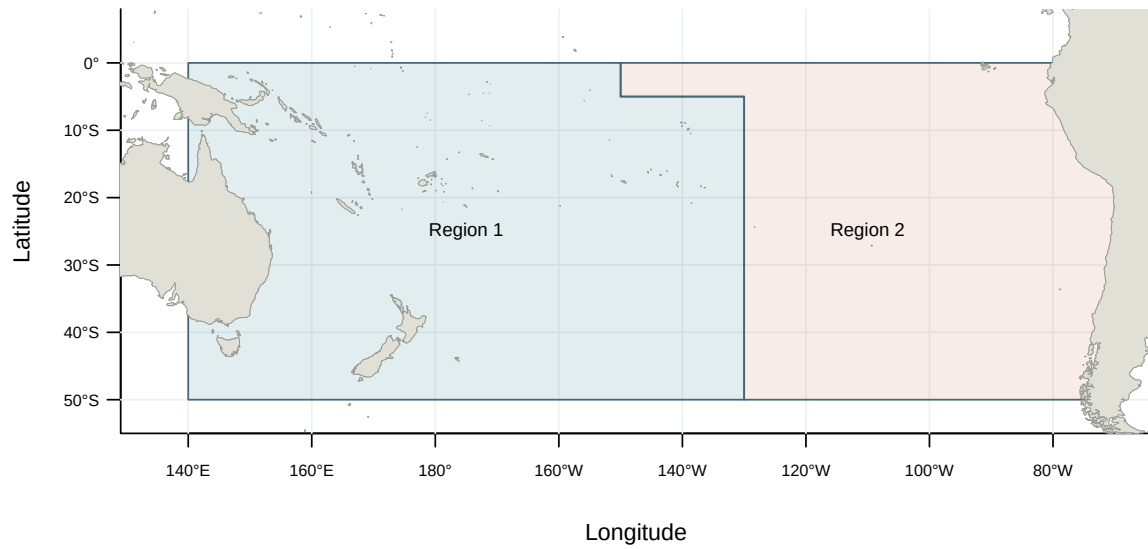


Figure 1: Model regions for the 2024 South Pacific albacore assessment (WCPFC-SC20-SA-WP-02, Rev.03, Figure 2).

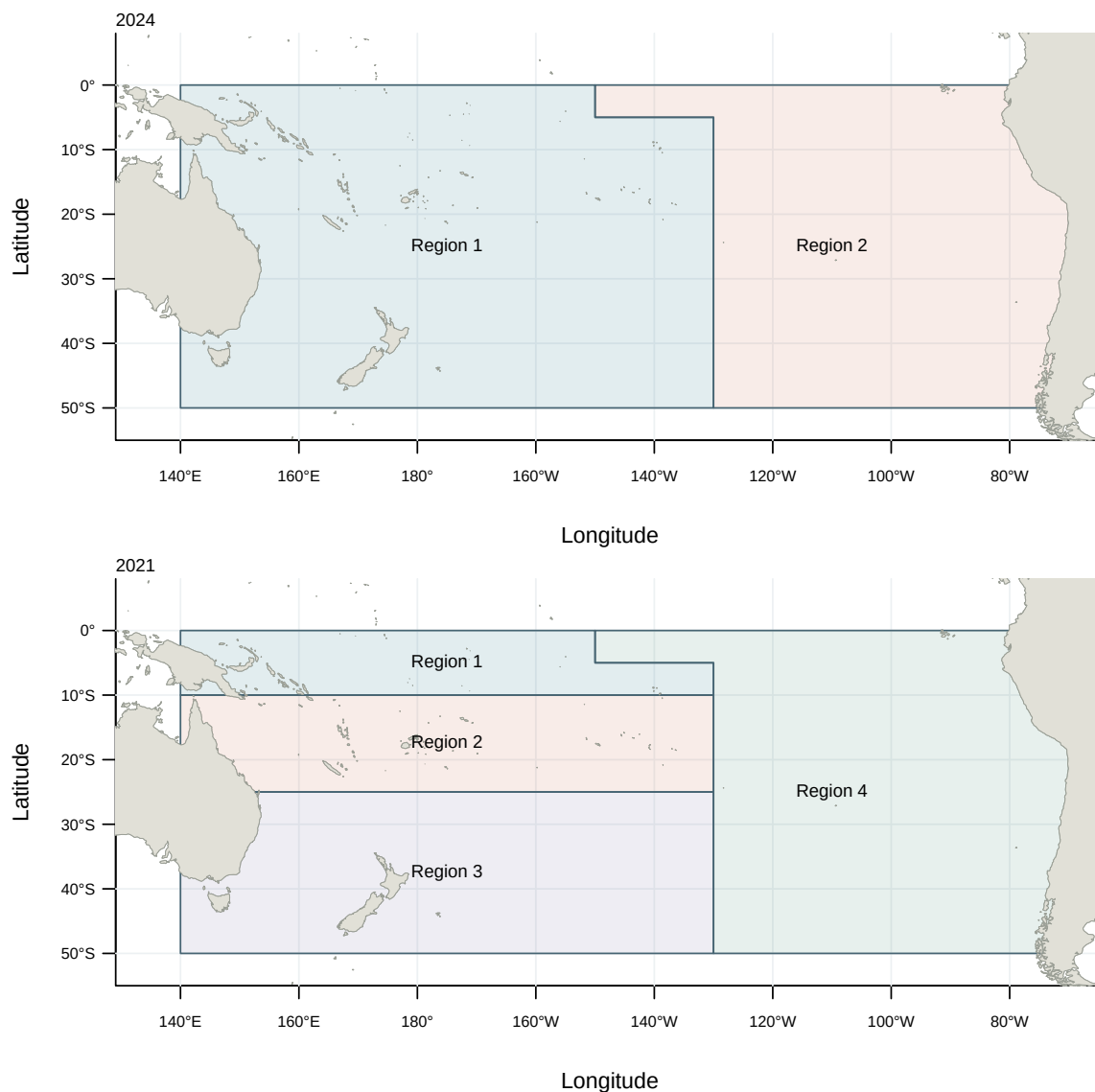


Figure 2: Spatial structures of the 2024 and 2021 South Pacific albacore assessments (WCPFC-SC20-SA-WP-02, Rev.03, Figures 1–2).



Figure 3: Illustrative output. Catch by fishing group.

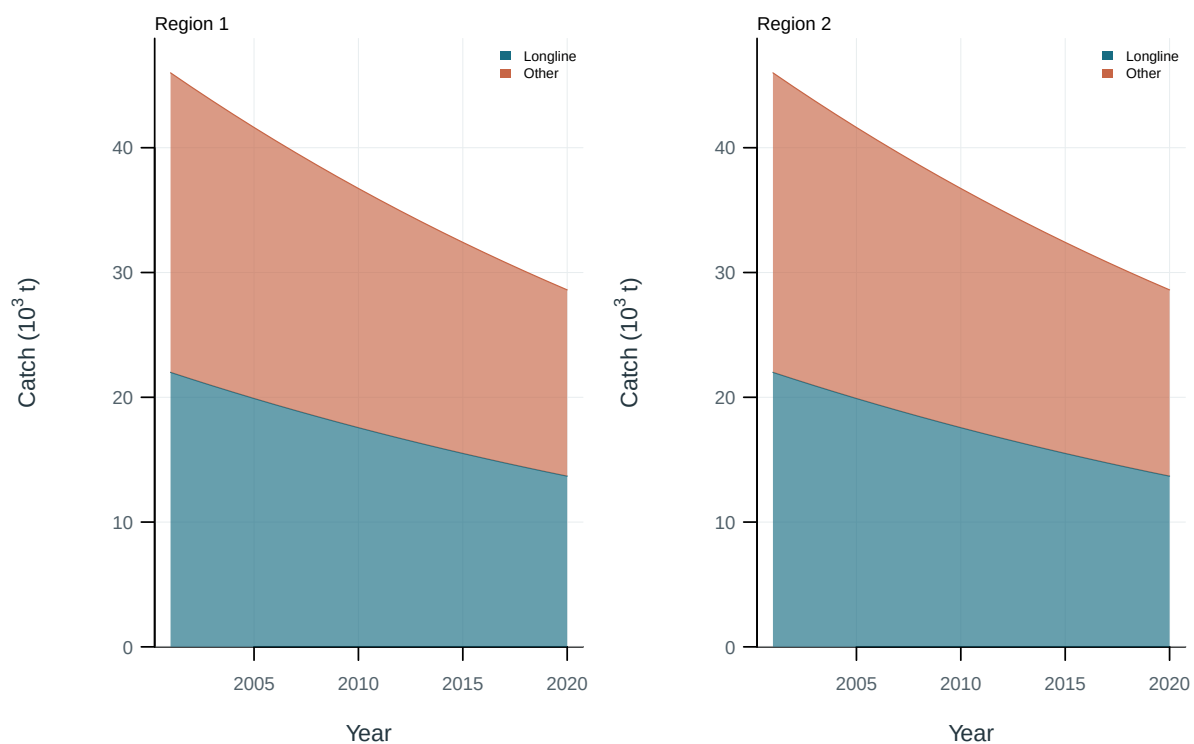


Figure 4: Illustrative output. Catch by fishing group and region.

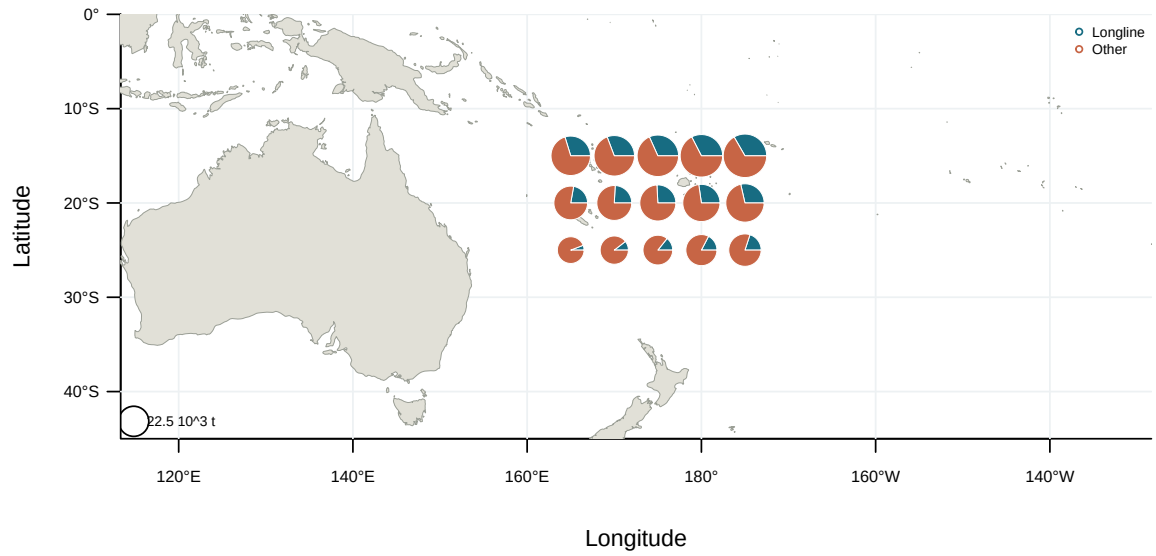


Figure 5: Illustrative output. Spatial catch distribution by fishing group.



Figure 6: Observation availability by data type and fishery. Coloured cells indicate years with observations.

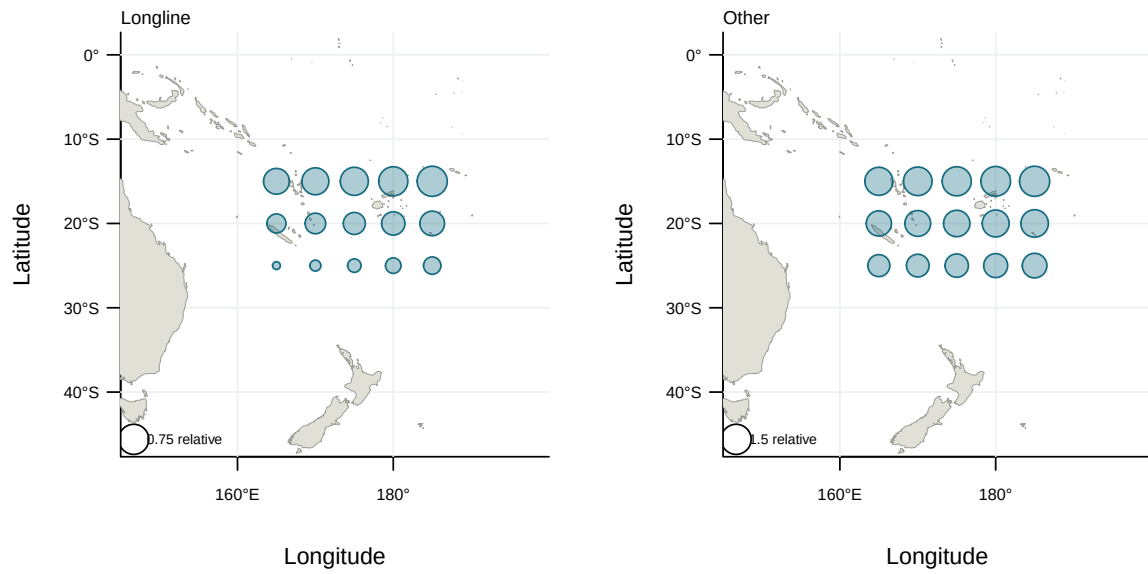


Figure 7: Illustrative output. Spatial distribution of nominal abundance indices.

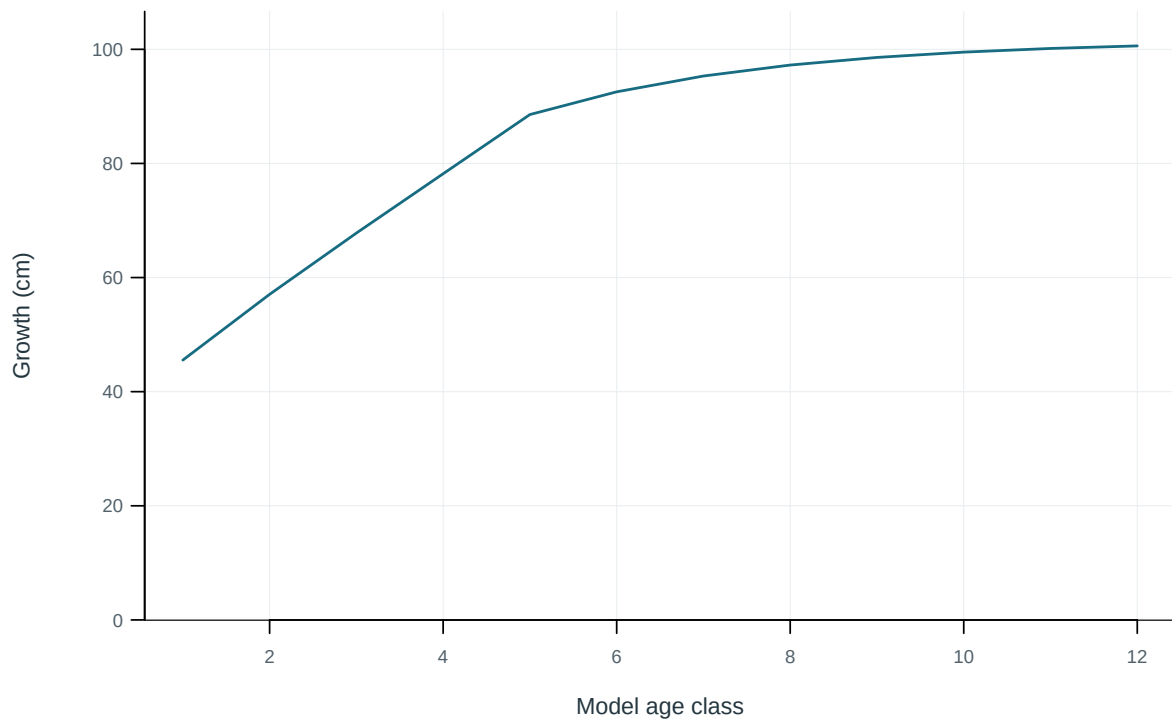


Figure 8: Mean length at the start of each model age class.

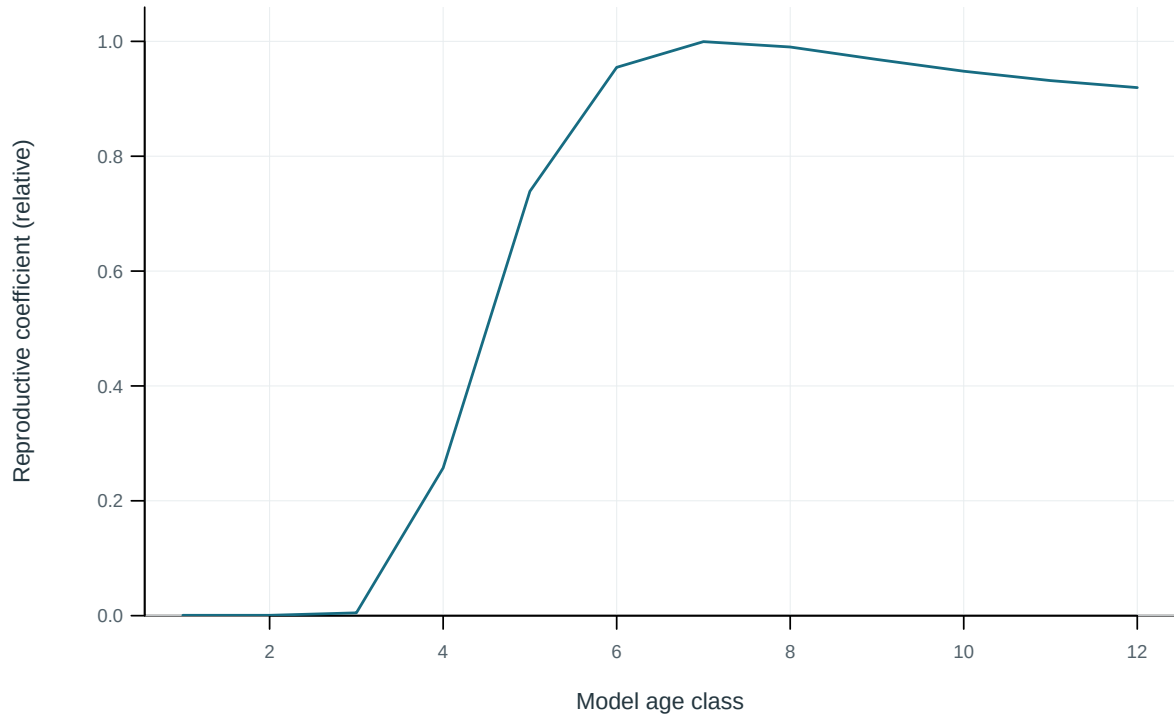


Figure 9: Model reproductive coefficient at age.

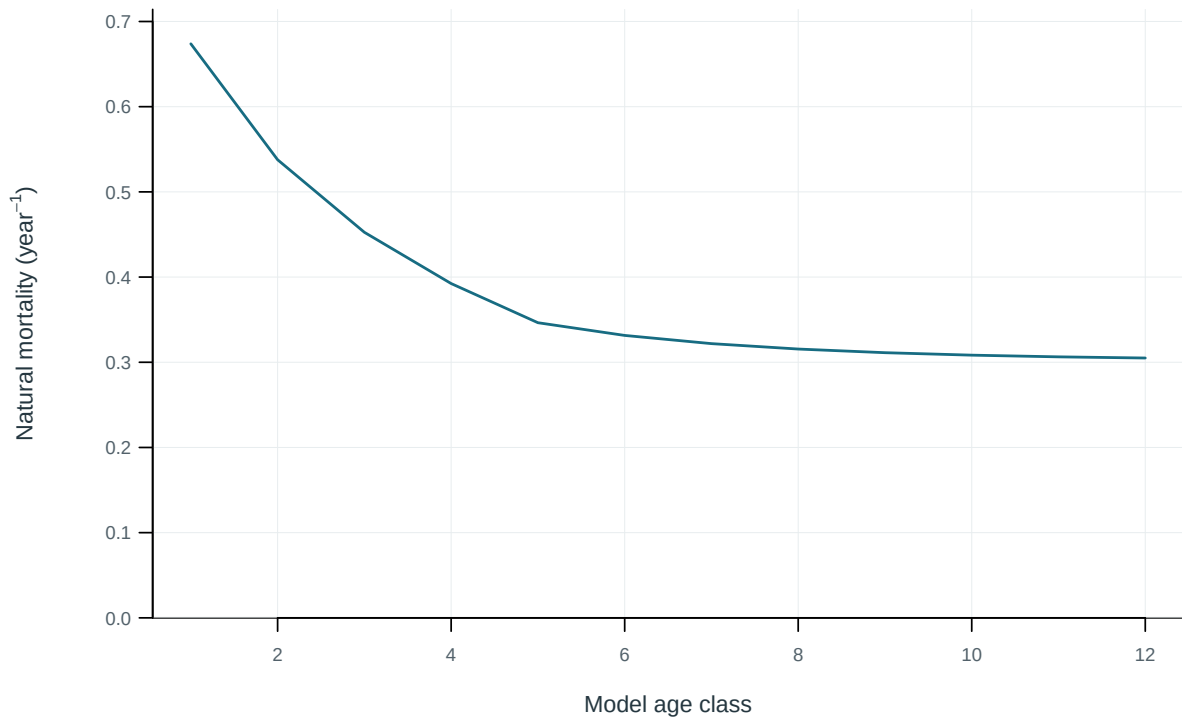


Figure 10: Annual instantaneous natural mortality at age.

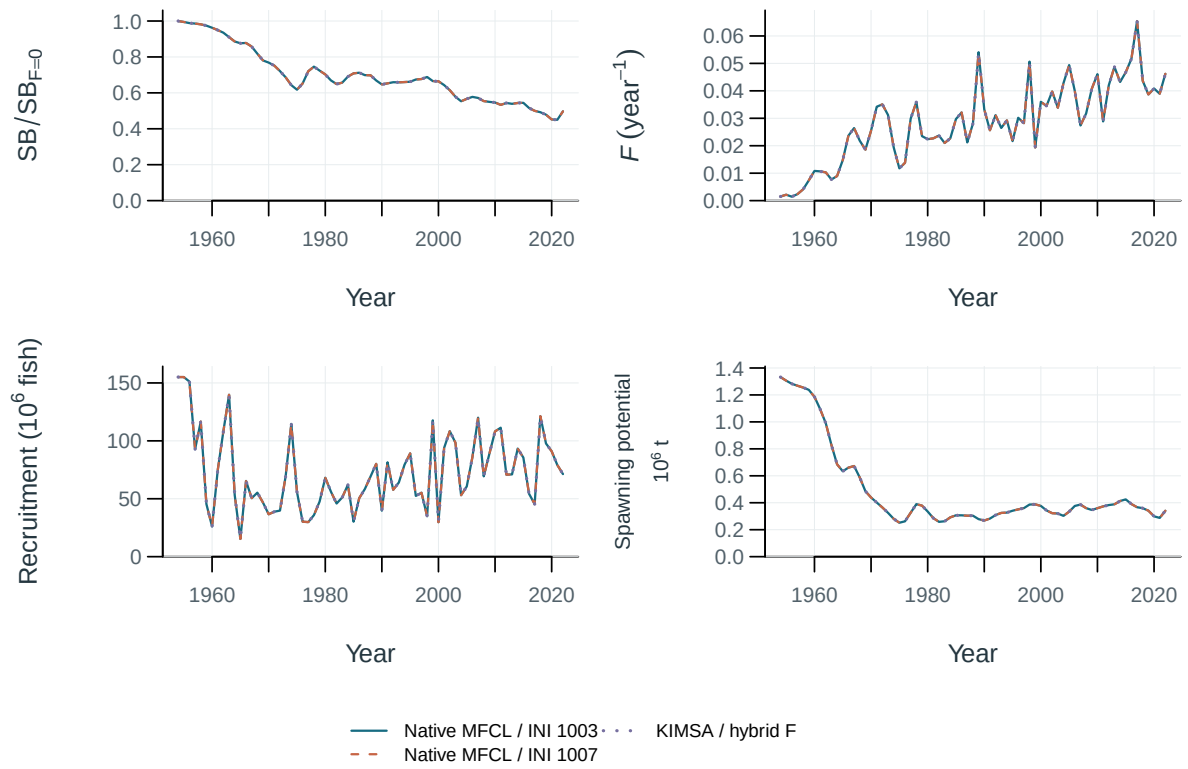


Figure 11: Stepwise model trajectories.

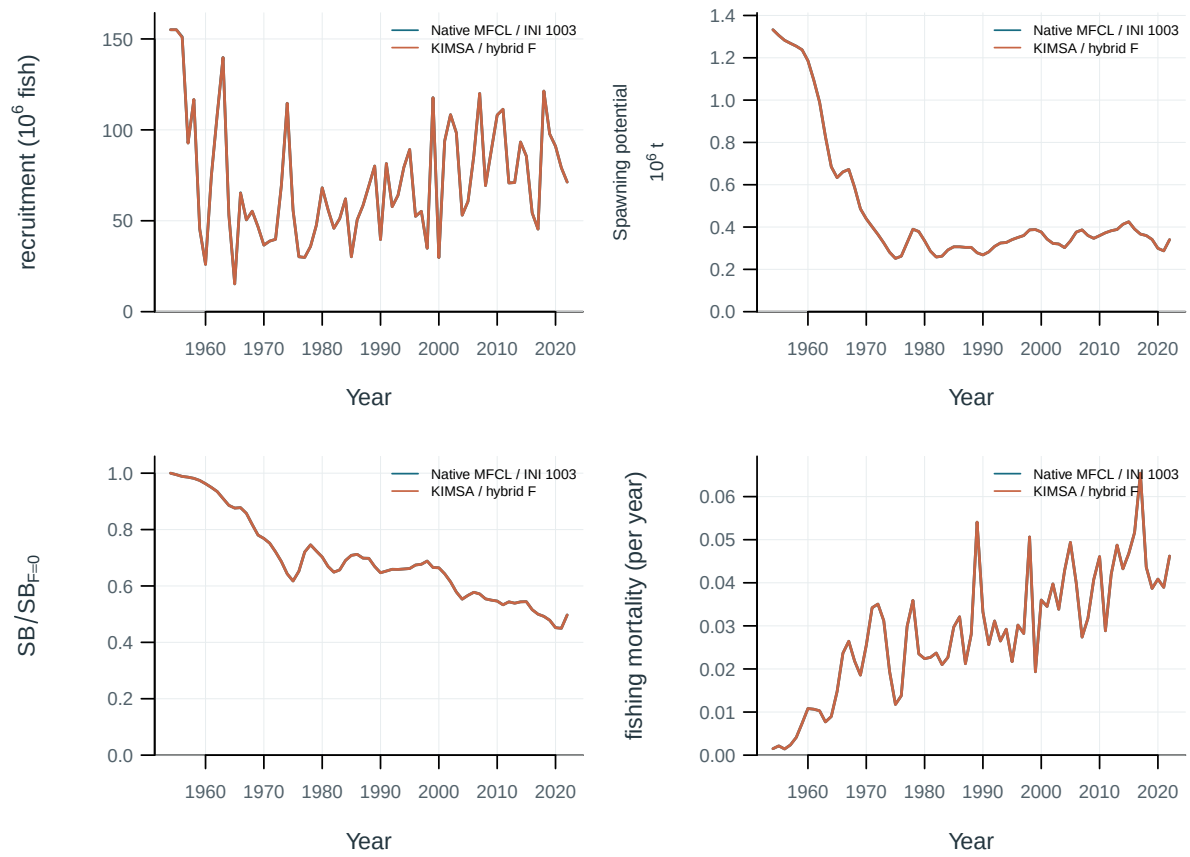


Figure 12: Selected model-development trajectories.

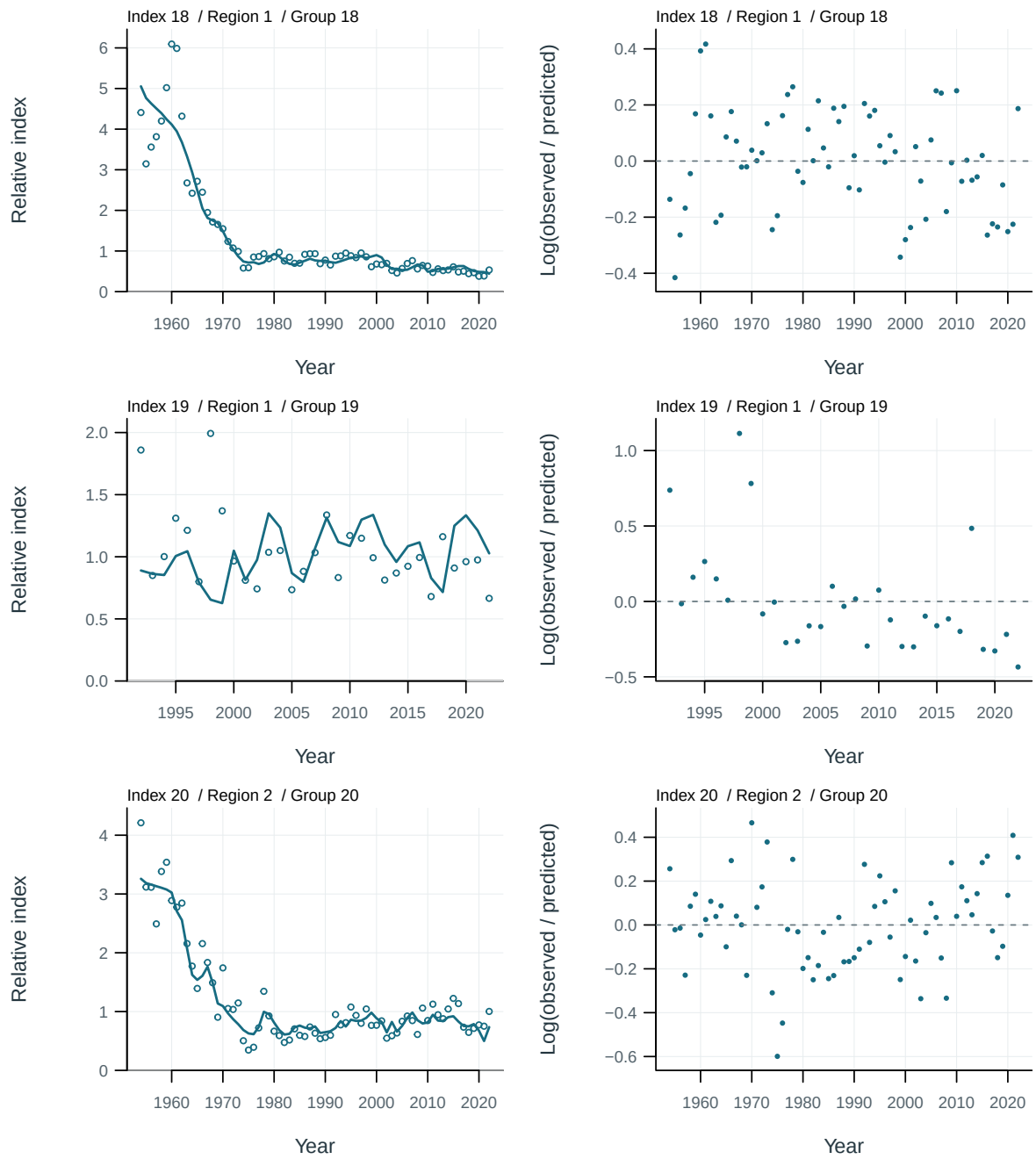


Figure 13: Abundance-index fits and log residuals by series. Residuals are $\log(\text{observed}/\text{predicted})$.

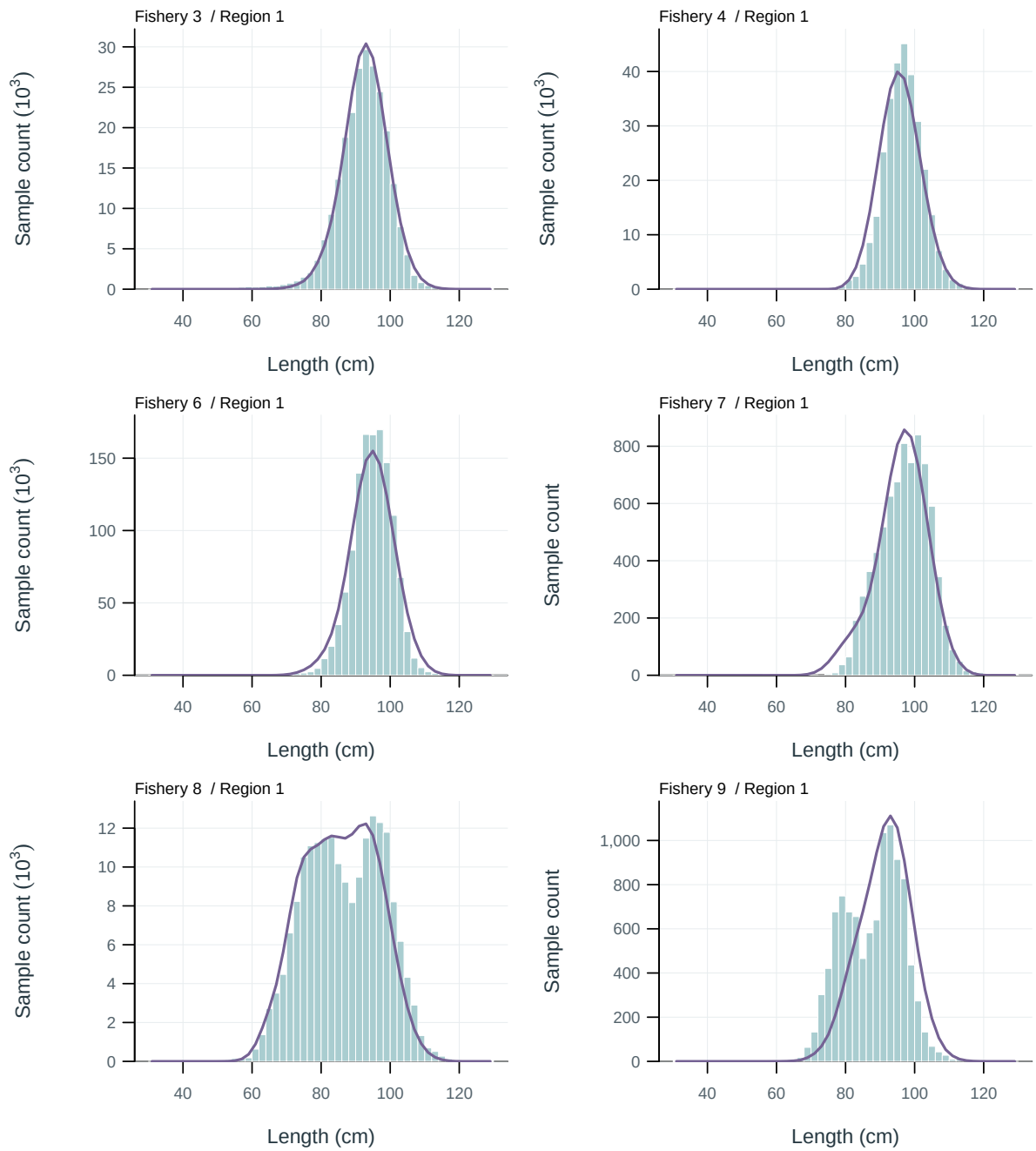


Figure 14: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

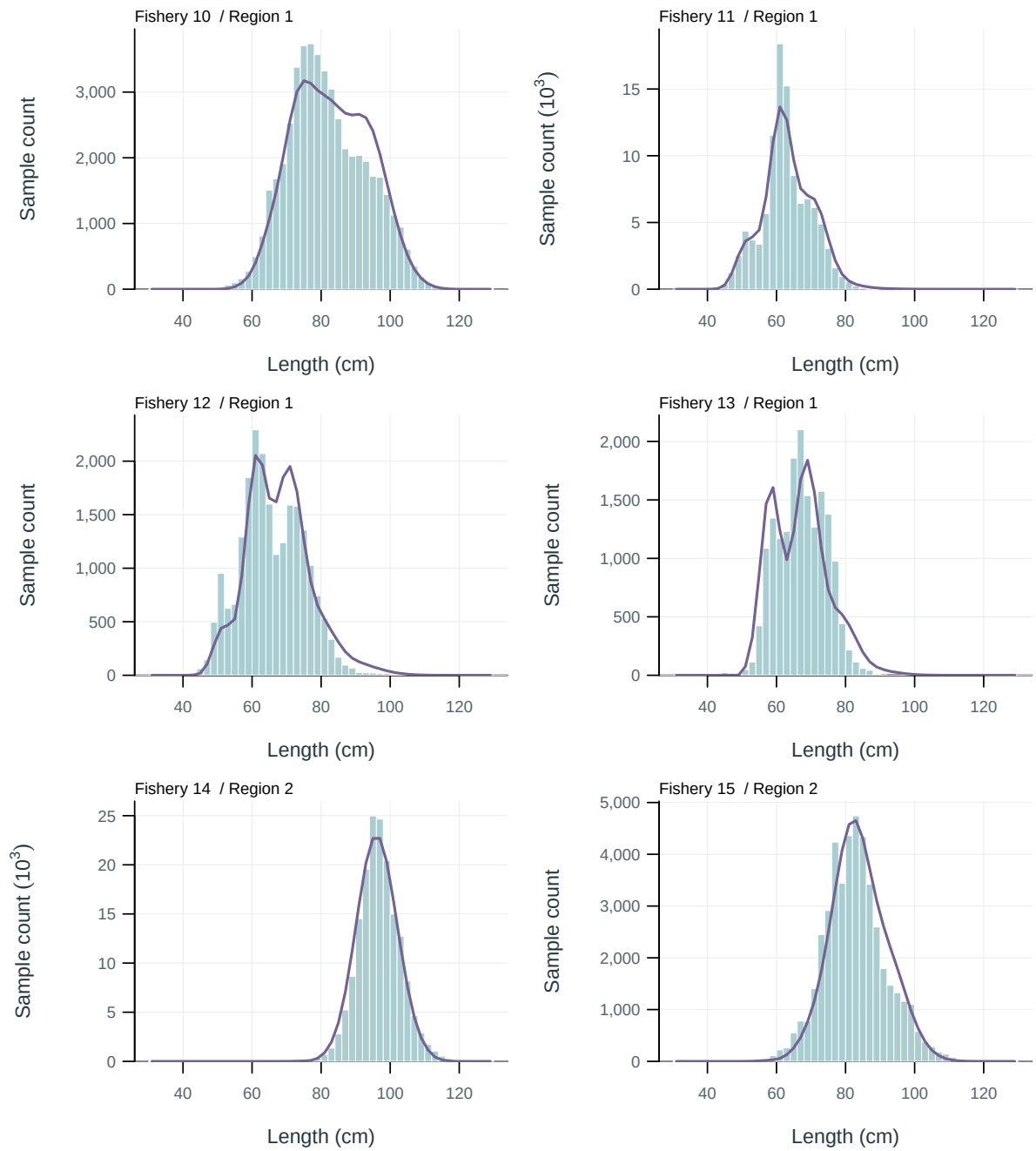


Figure 15: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

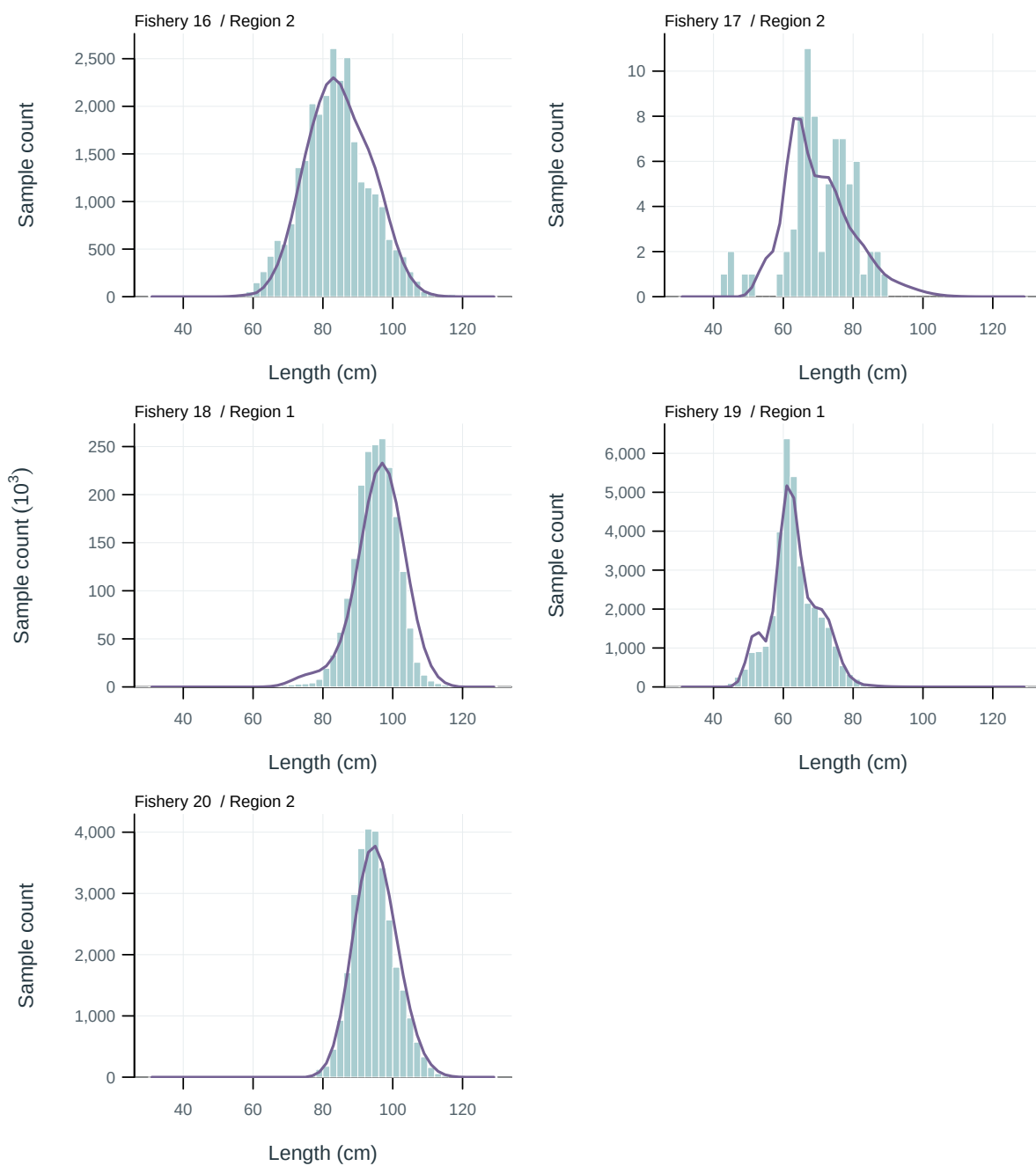


Figure 16: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

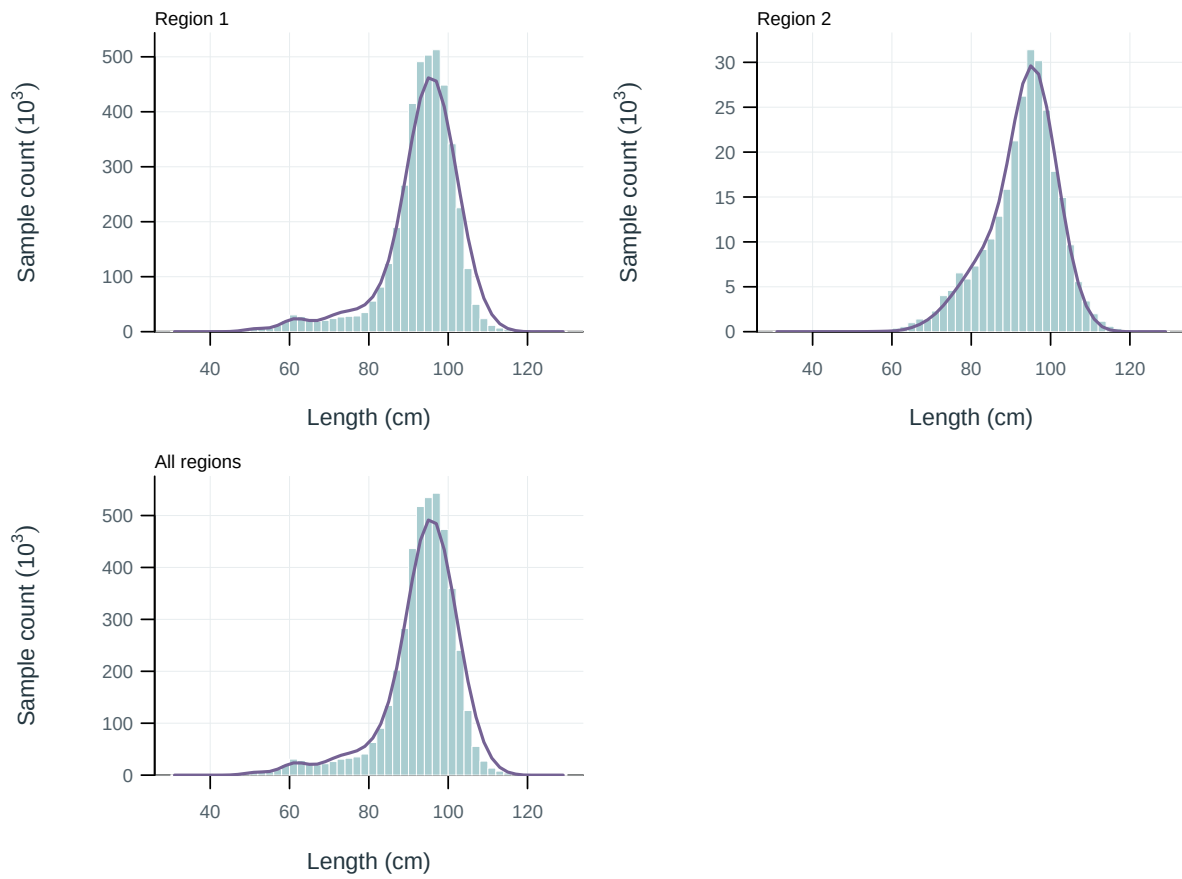


Figure 17: Length-composition fits by region. Observed sample counts (bars) and predictions (lines), pooled over years.

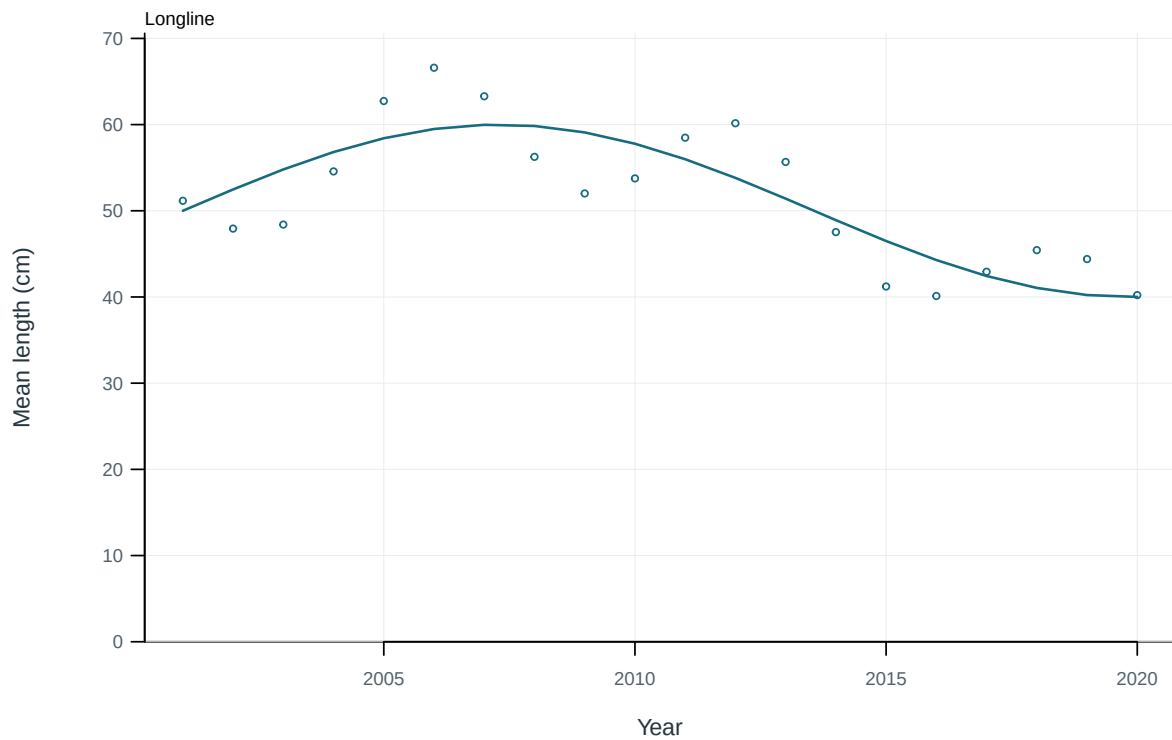


Figure 18: Illustrative output. Observed and predicted length summaries for longline fisheries.



Figure 19: Illustrative output. Observed and predicted length summaries for other fisheries.

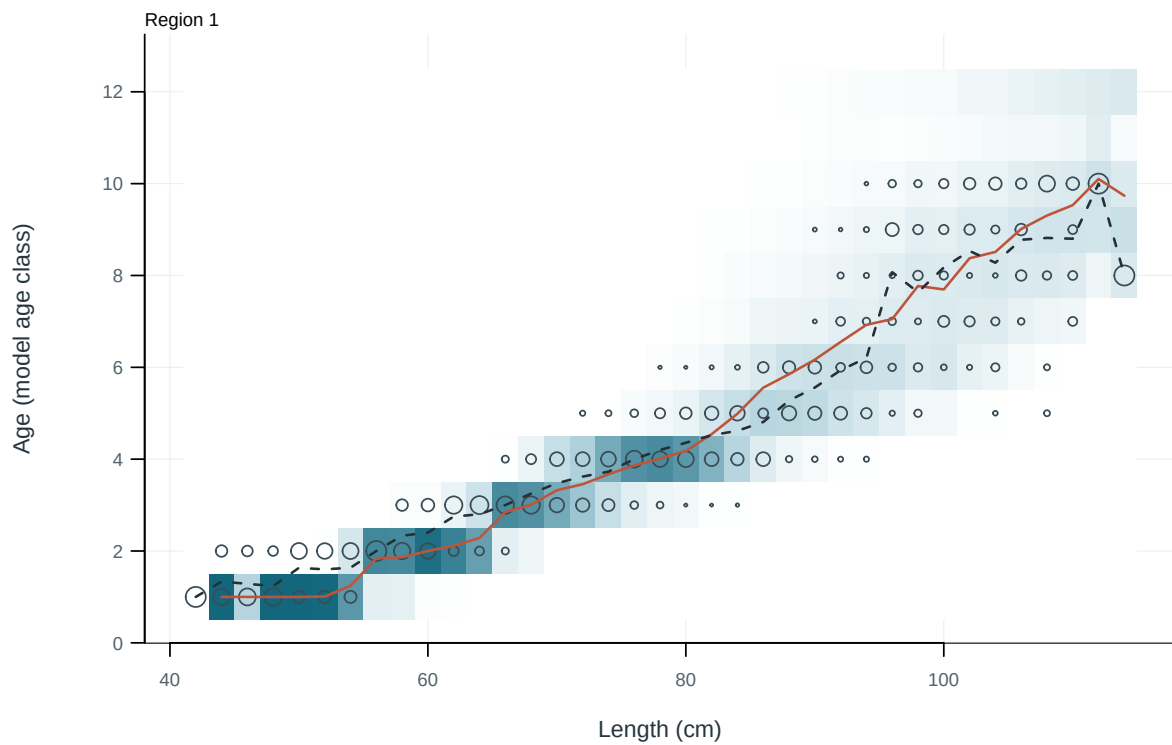


Figure 20: Observed and predicted age distributions conditional on length. Cells: predictions; circles: observations; solid and dashed lines: predicted and observed mean age.

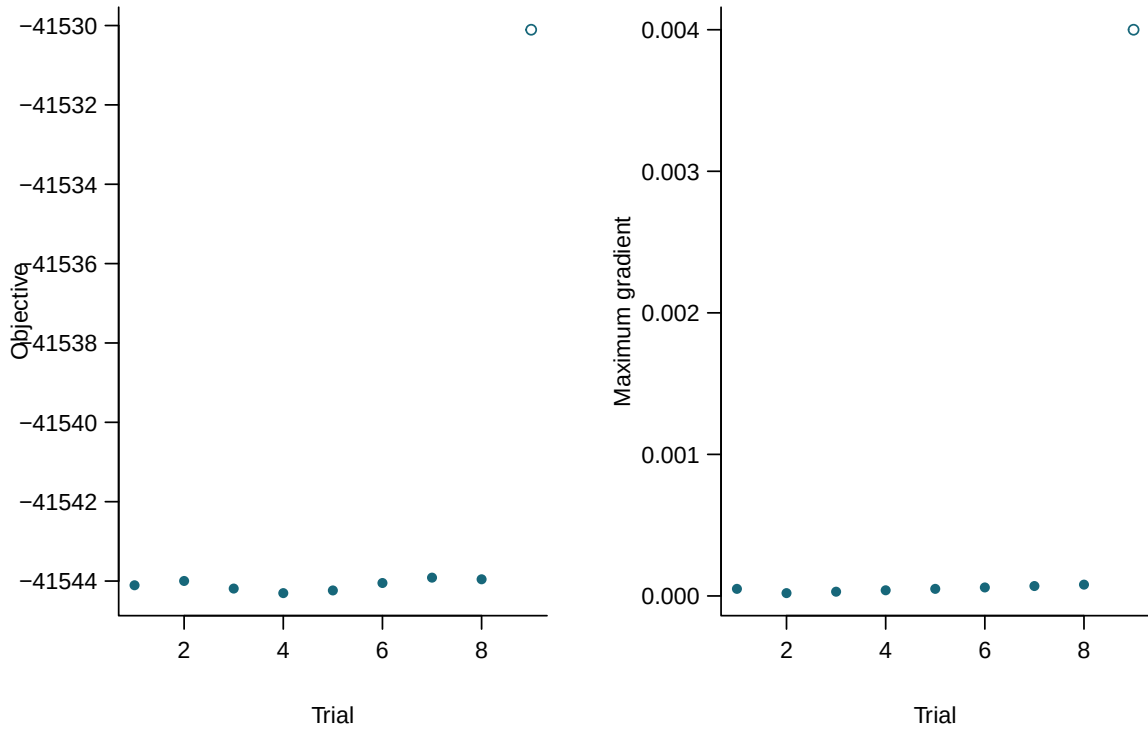


Figure 21: Final objectives and maximum gradients for independent starting values. Filled points meet all recorded numerical checks.

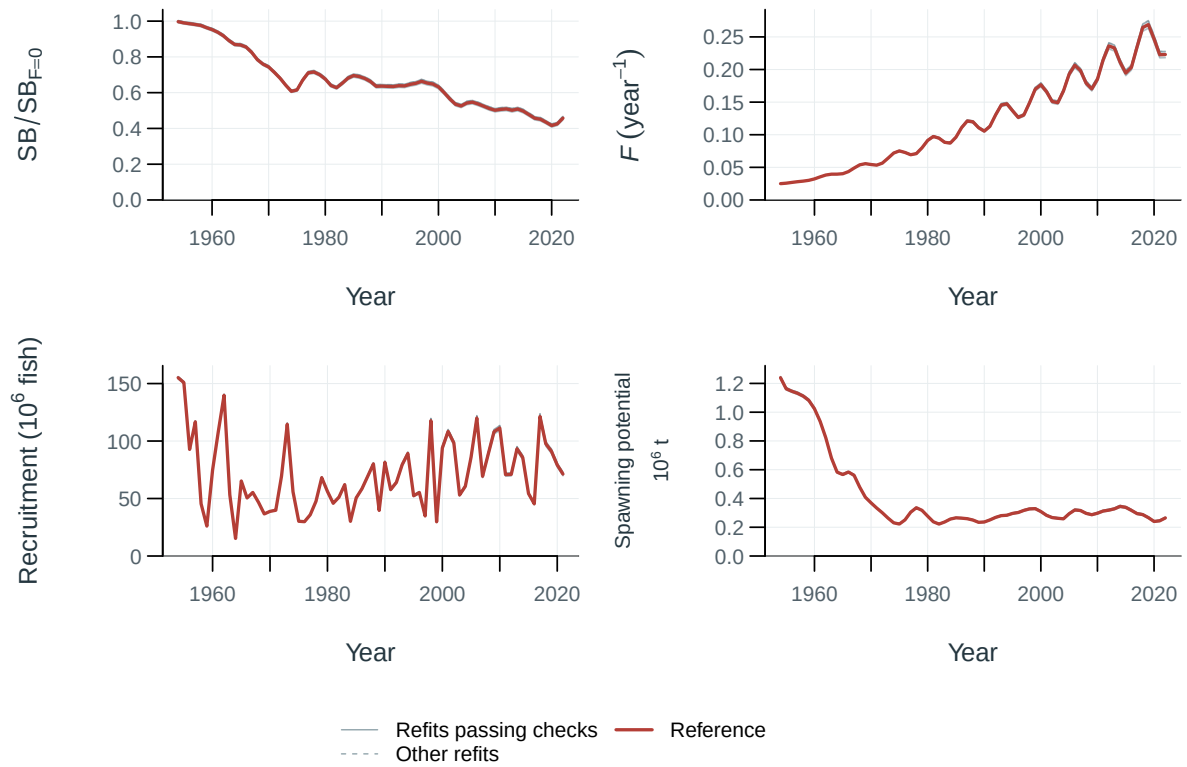


Figure 22: Jitter model trajectories.

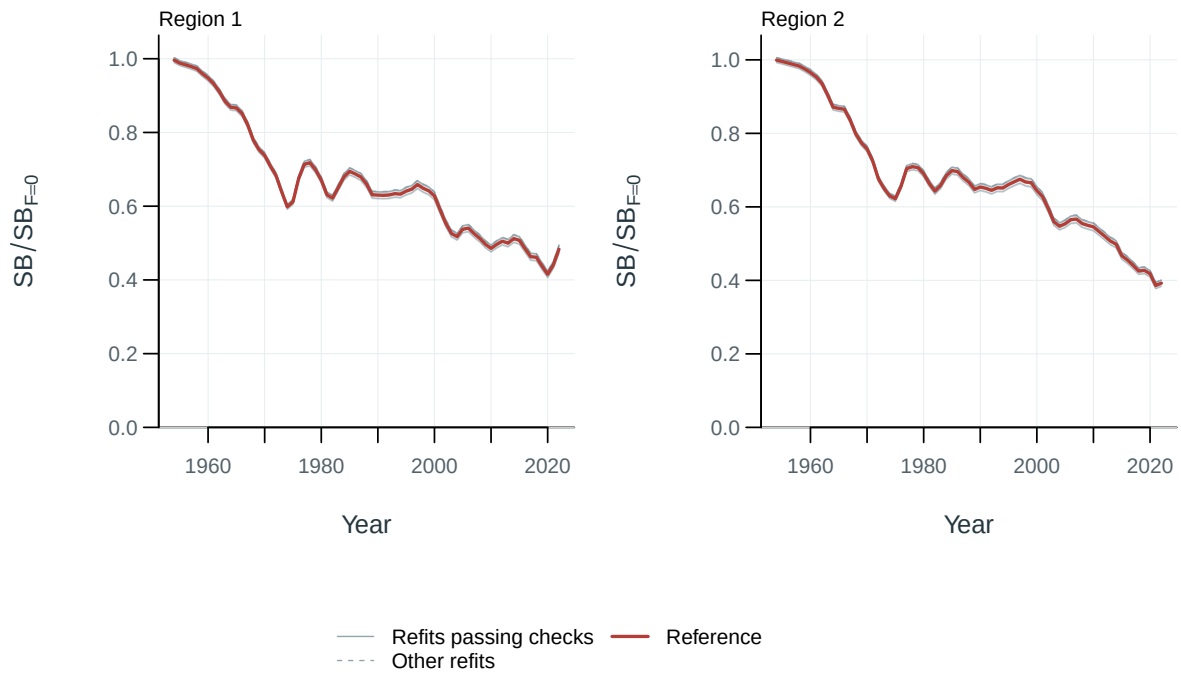


Figure 23: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

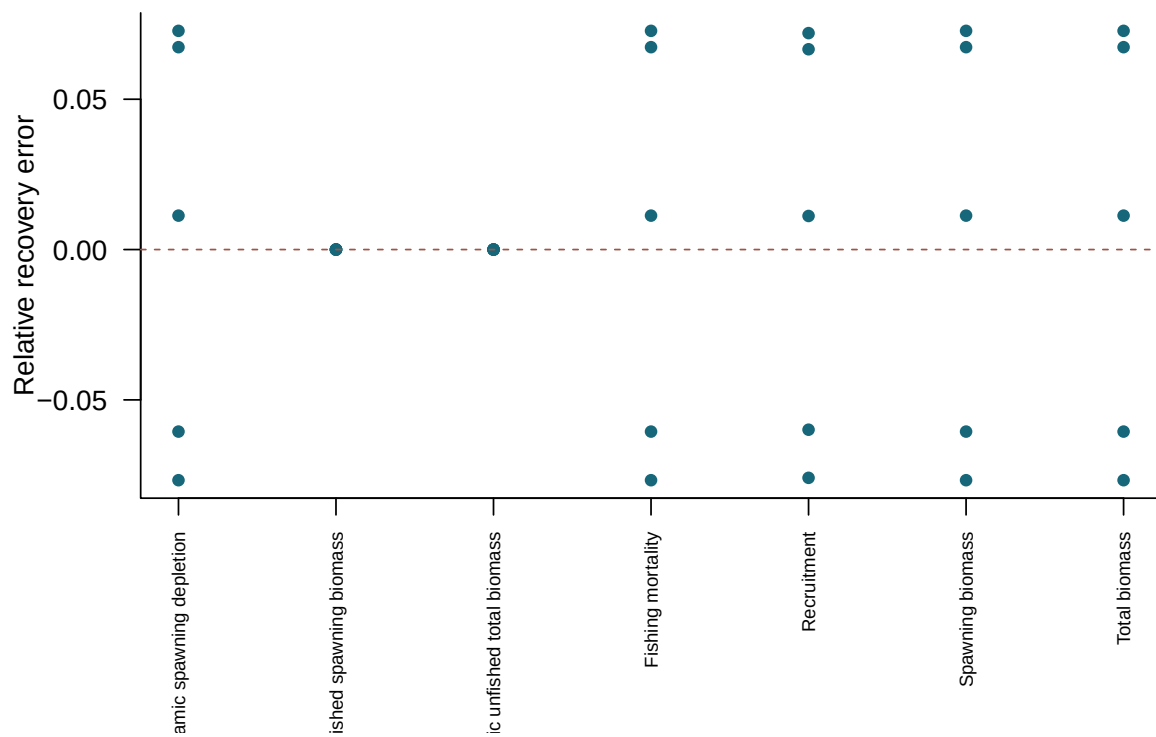


Figure 24: Relative recovery error in terminal quantities for converged simulated-data refits.

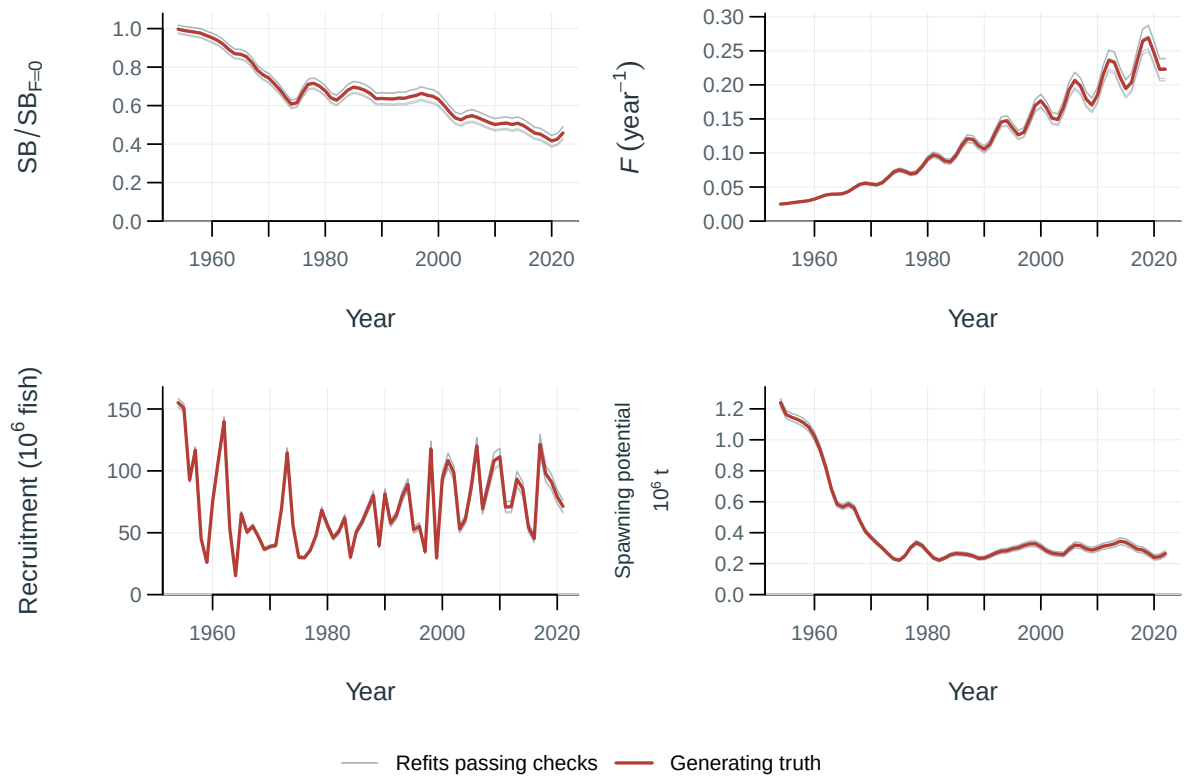


Figure 25: Selftest model trajectories.

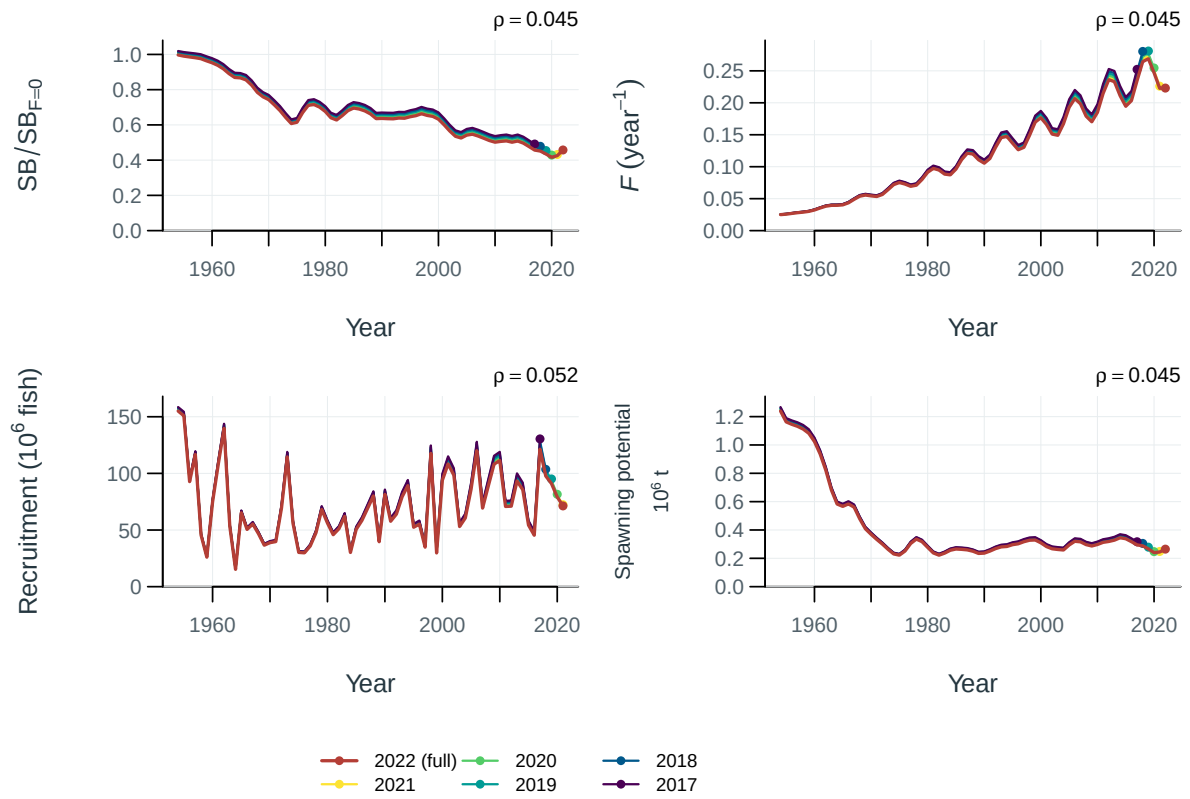


Figure 26: Population trajectories for the full-data fit and retrospective peels. Points mark terminal years; rho is Mohn's mean terminal relative difference for peels passing the recorded numerical checks.

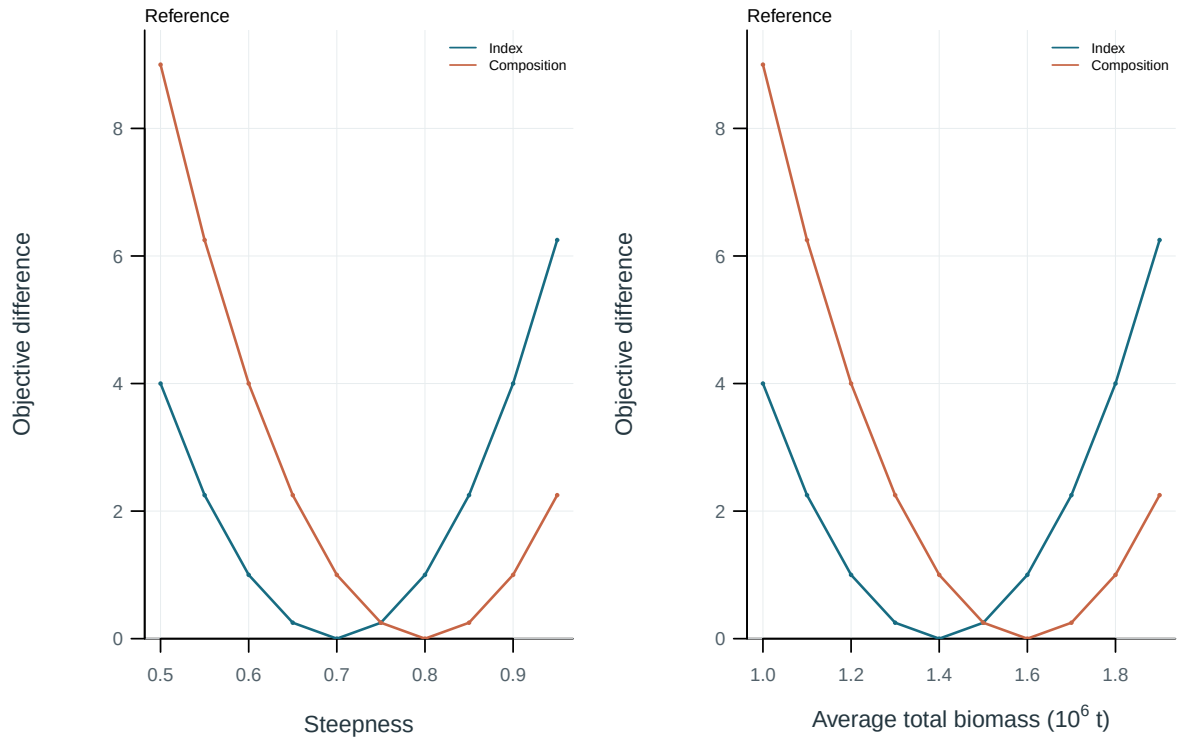


Figure 27: Total objective and component profiles. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

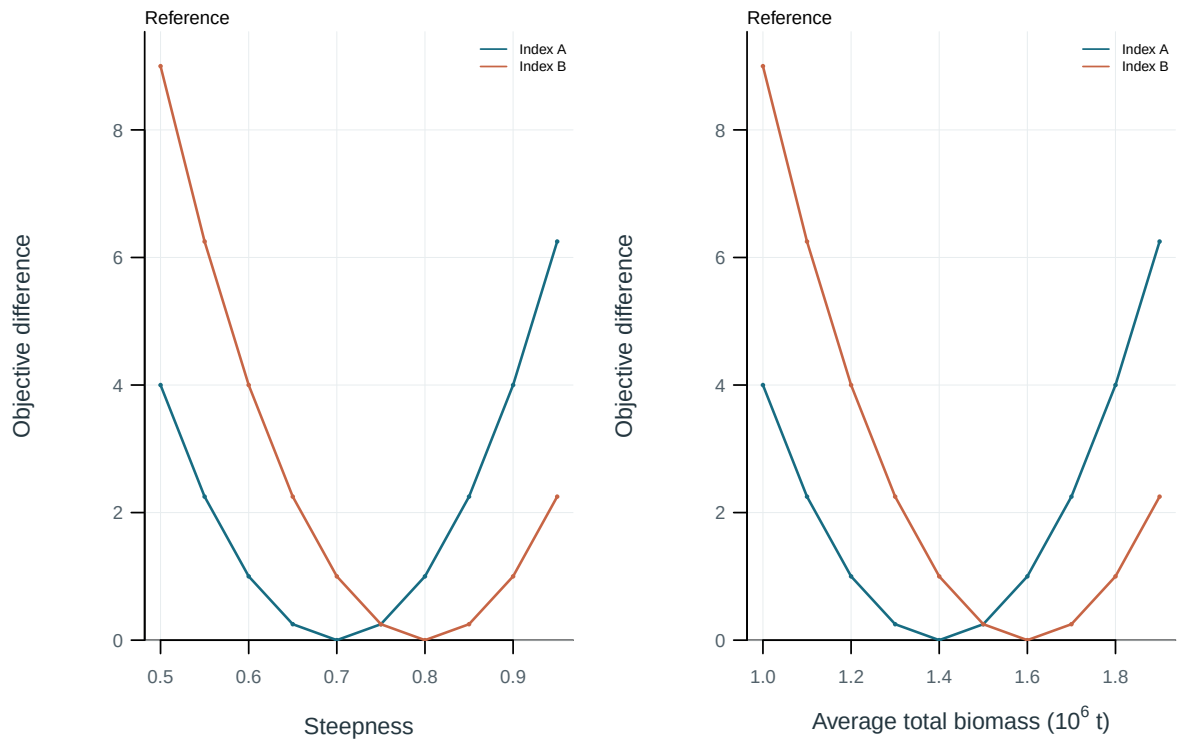


Figure 28: Objective profiles for abundance-index components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

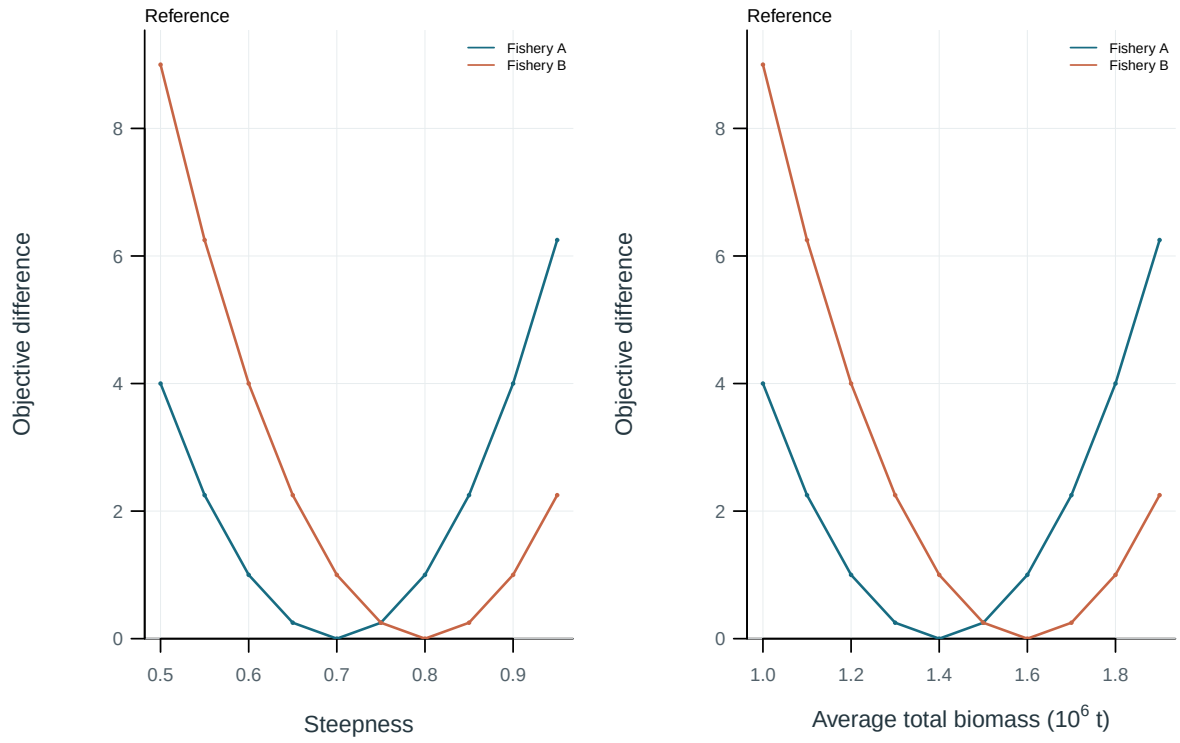


Figure 29: Objective profiles for length-composition components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

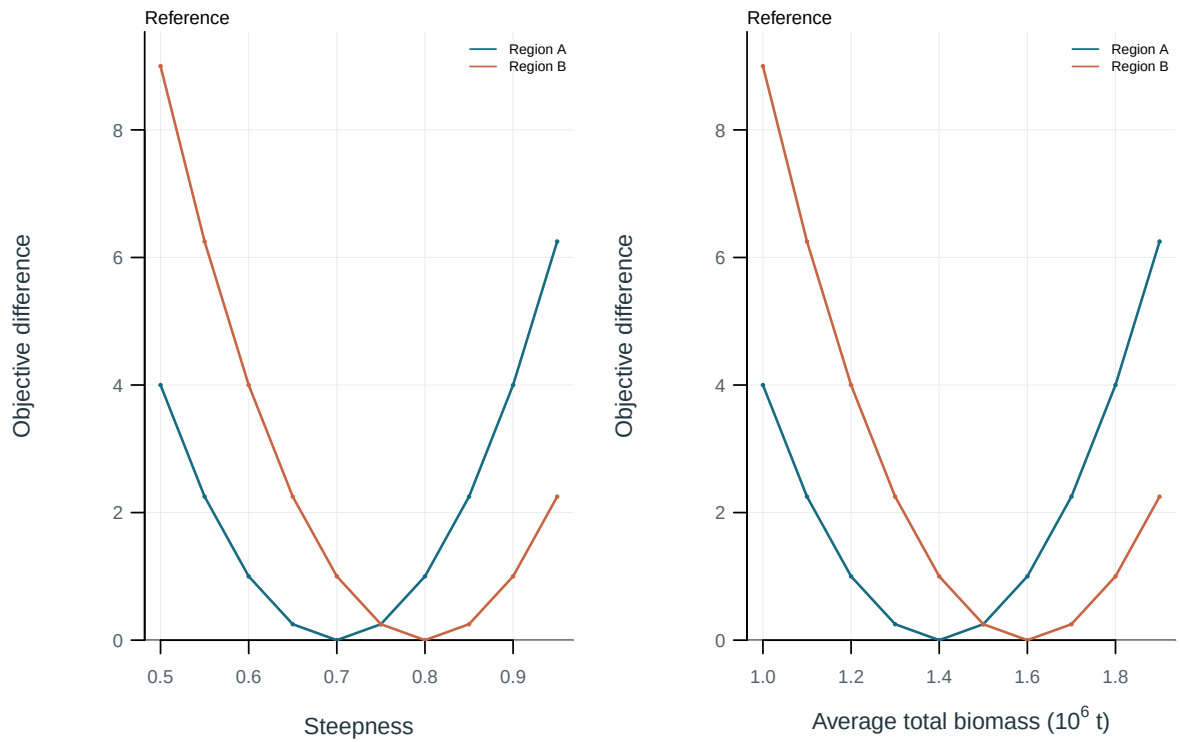


Figure 30: Objective profiles for conditional age-at-length components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

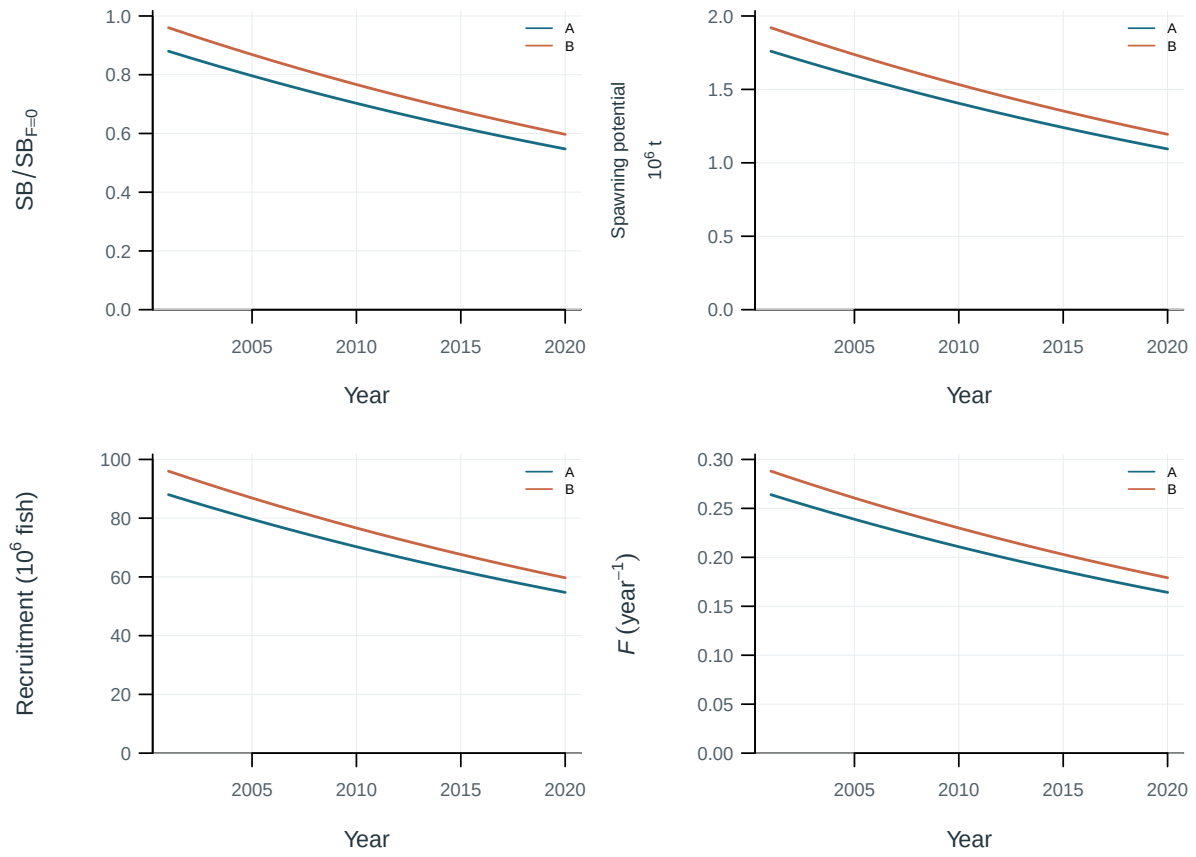


Figure 31: Illustrative output. Diagnostic and production-model trajectories.

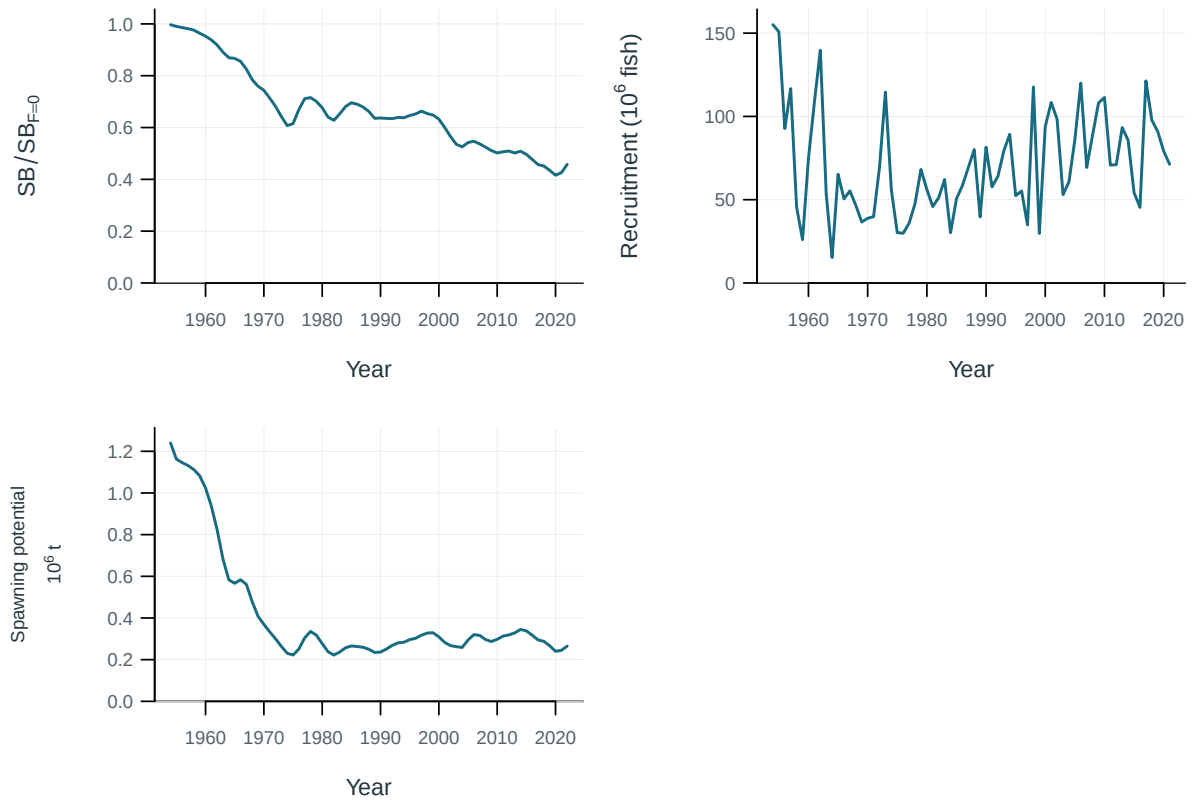


Figure 32: Estimated population trajectories.

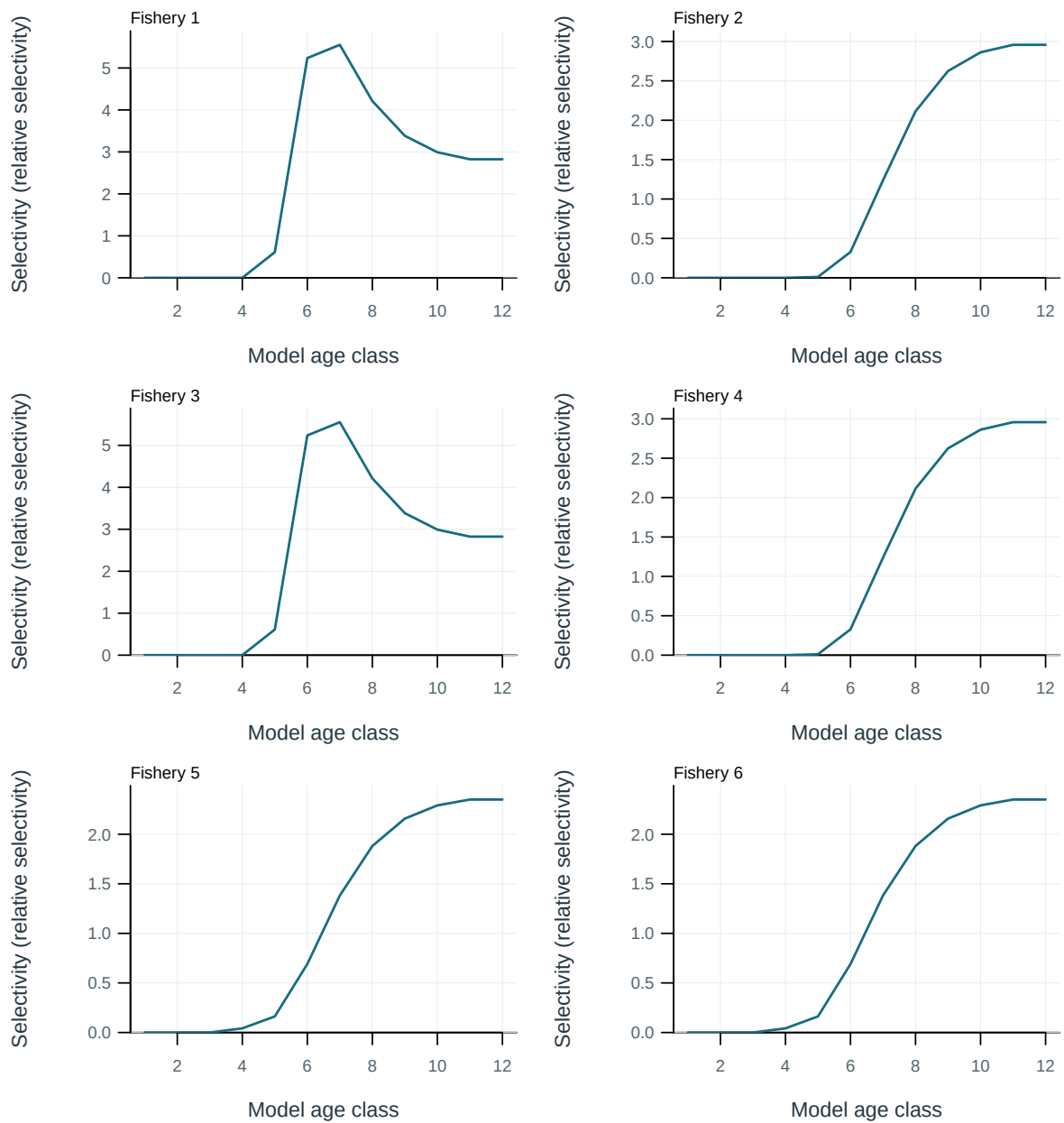


Figure 33: Fishery selectivity at age.

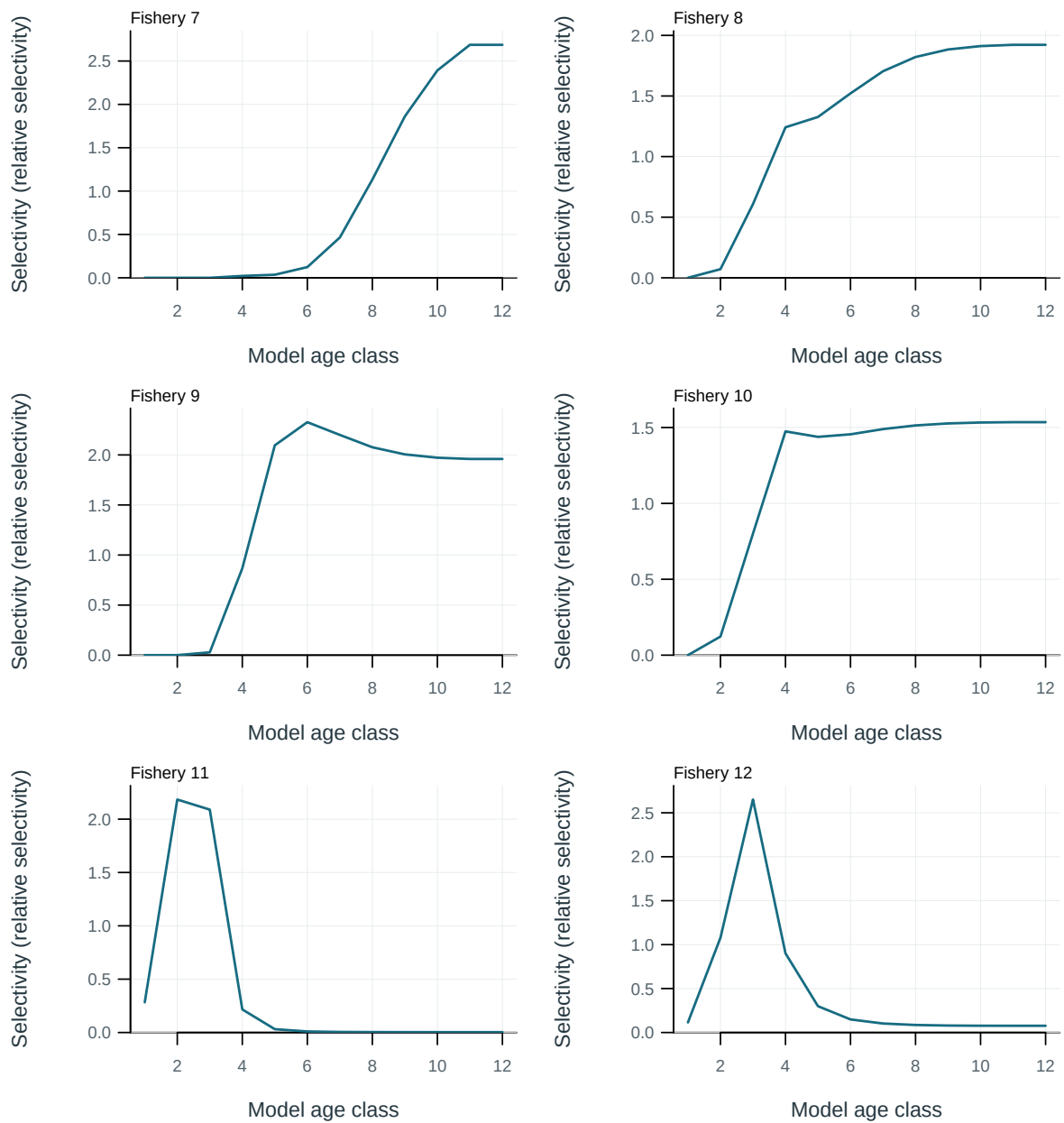


Figure 34: Fishery selectivity at age.

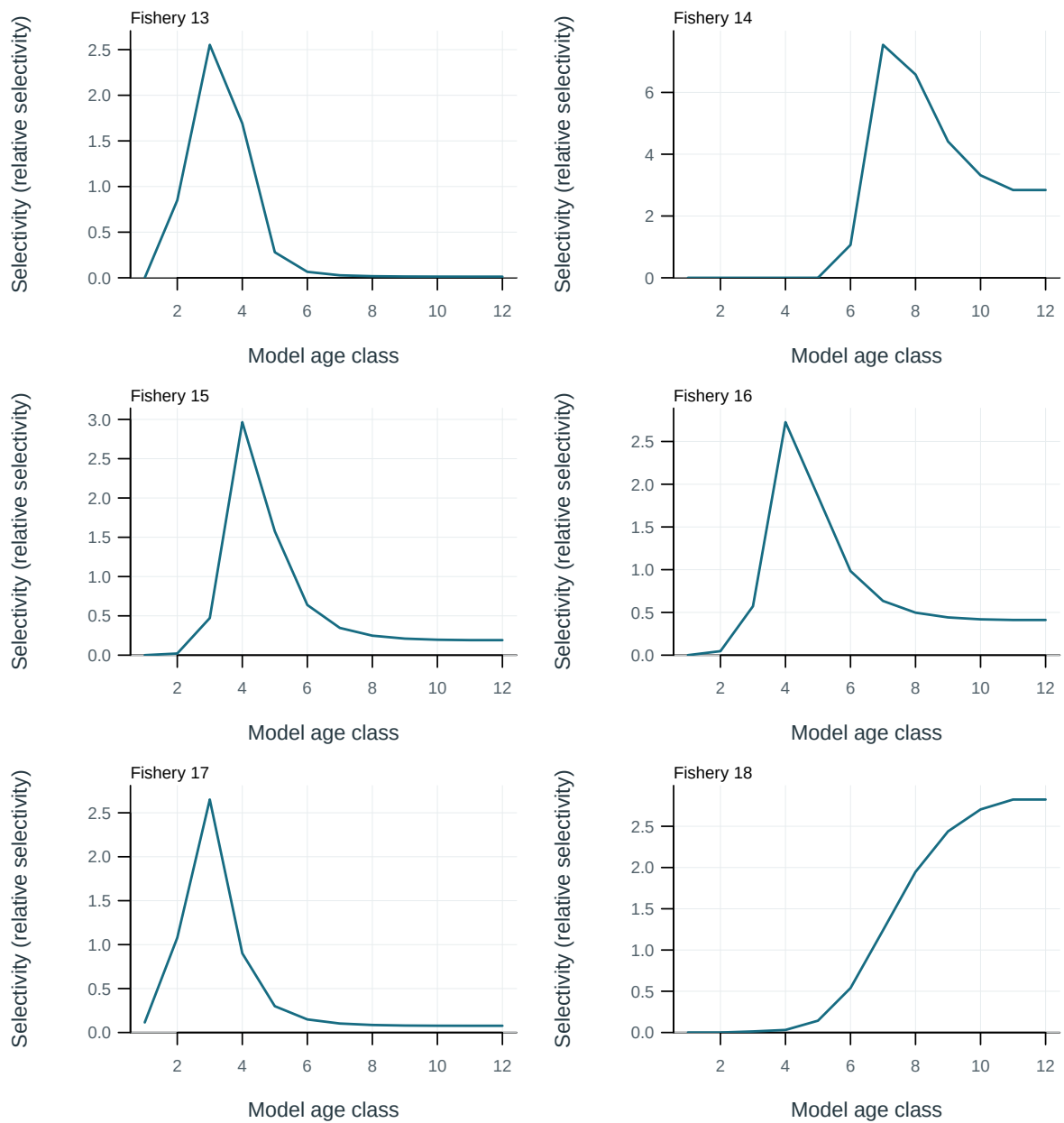


Figure 35: Fishery selectivity at age.

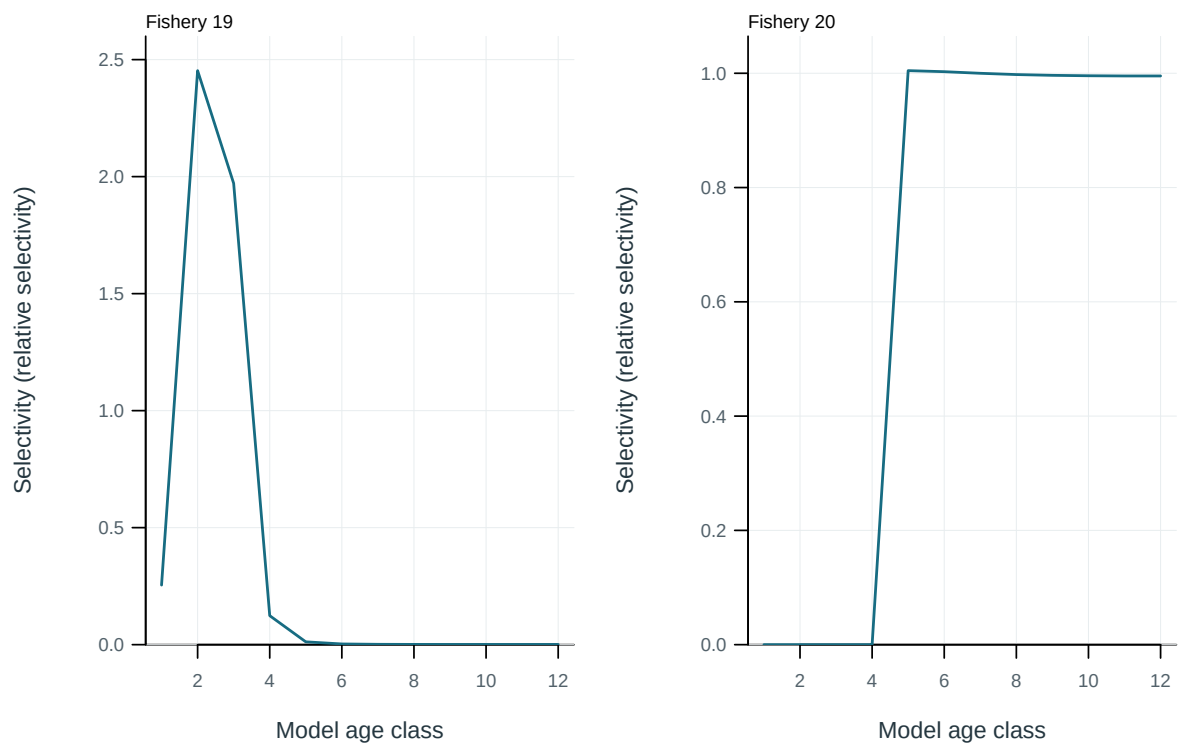


Figure 36: Fishery selectivity at age.

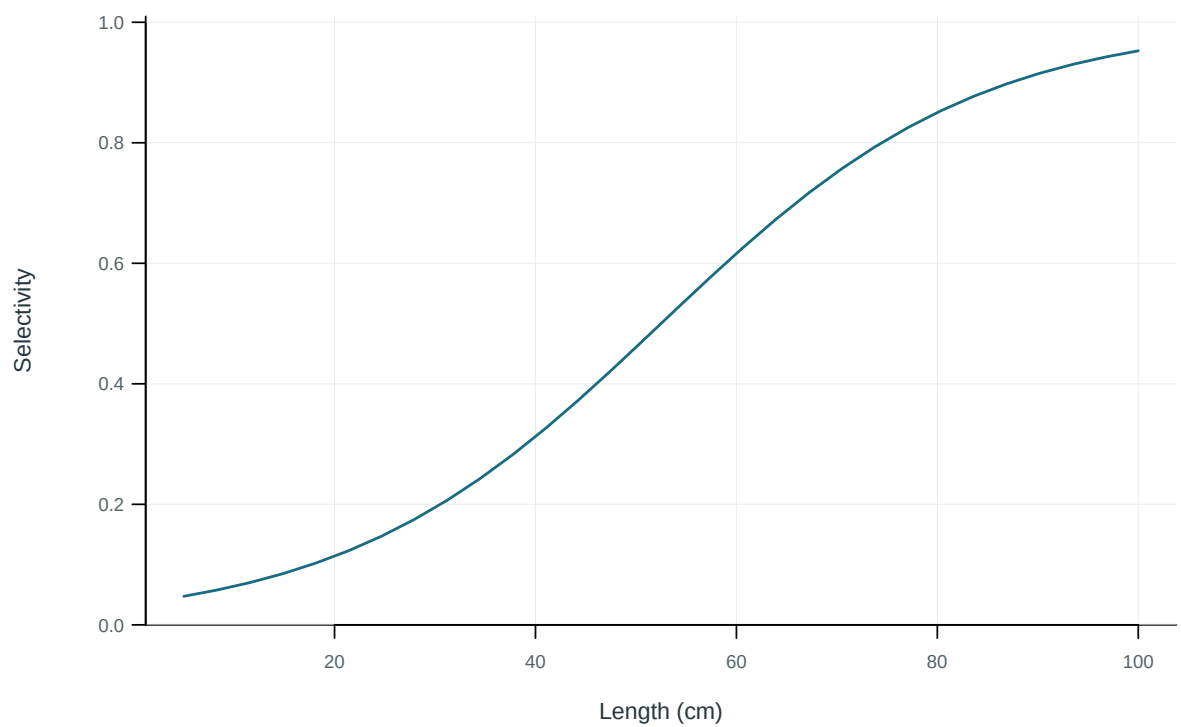


Figure 37: Illustrative output. Selectivity at length by fishery.

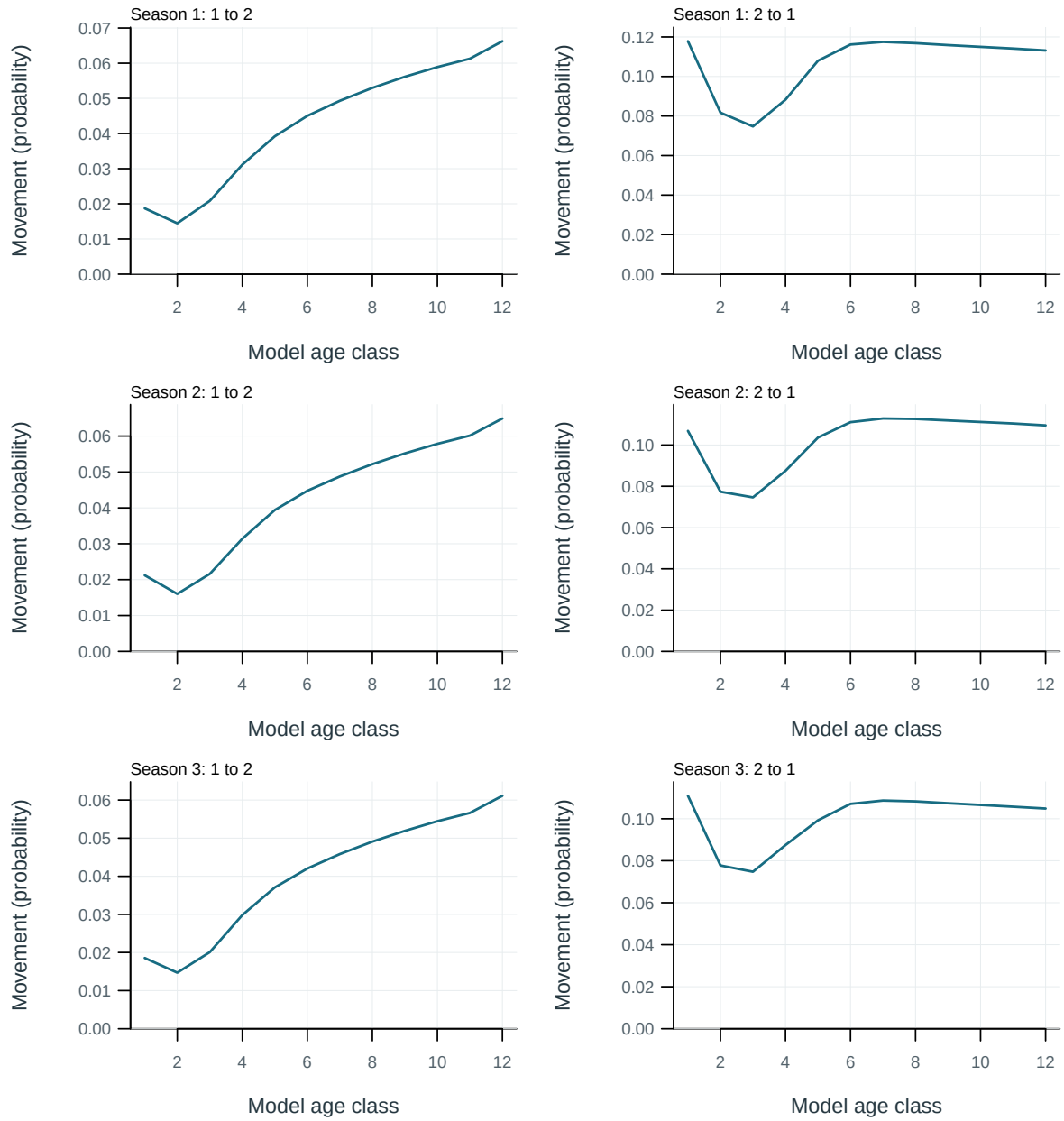


Figure 38: Probability of movement from source to destination region per movement event.

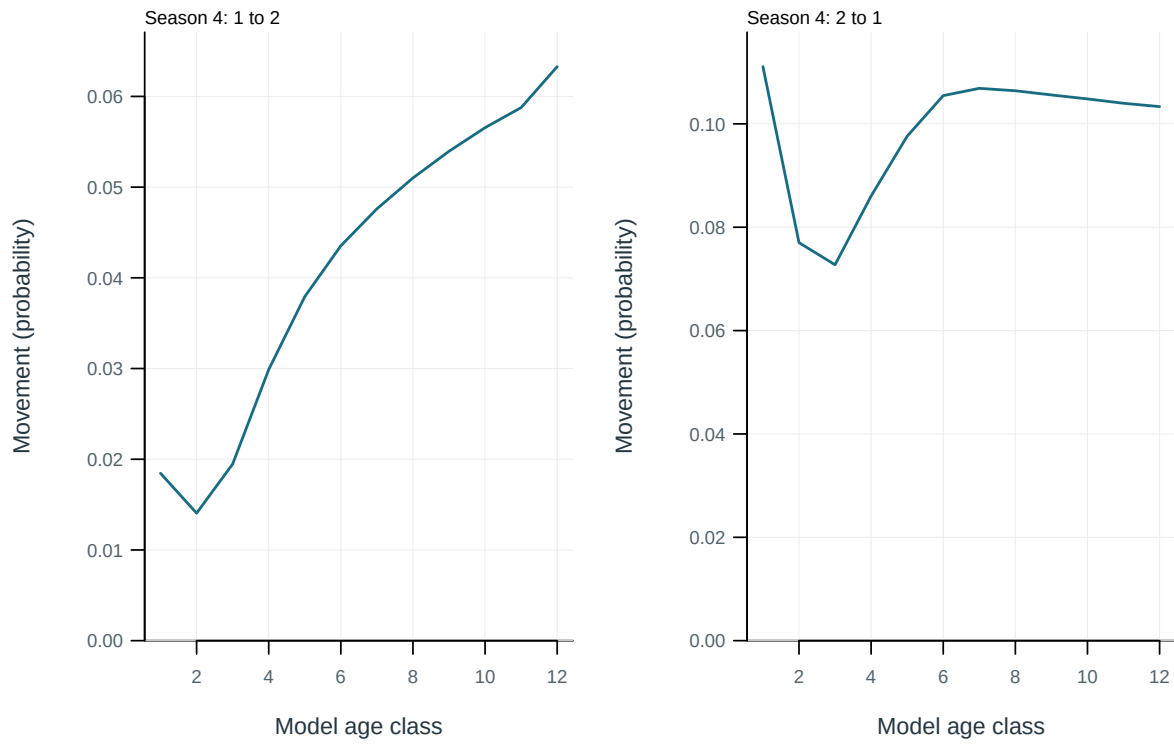


Figure 39: Probability of movement from source to destination region per movement event.

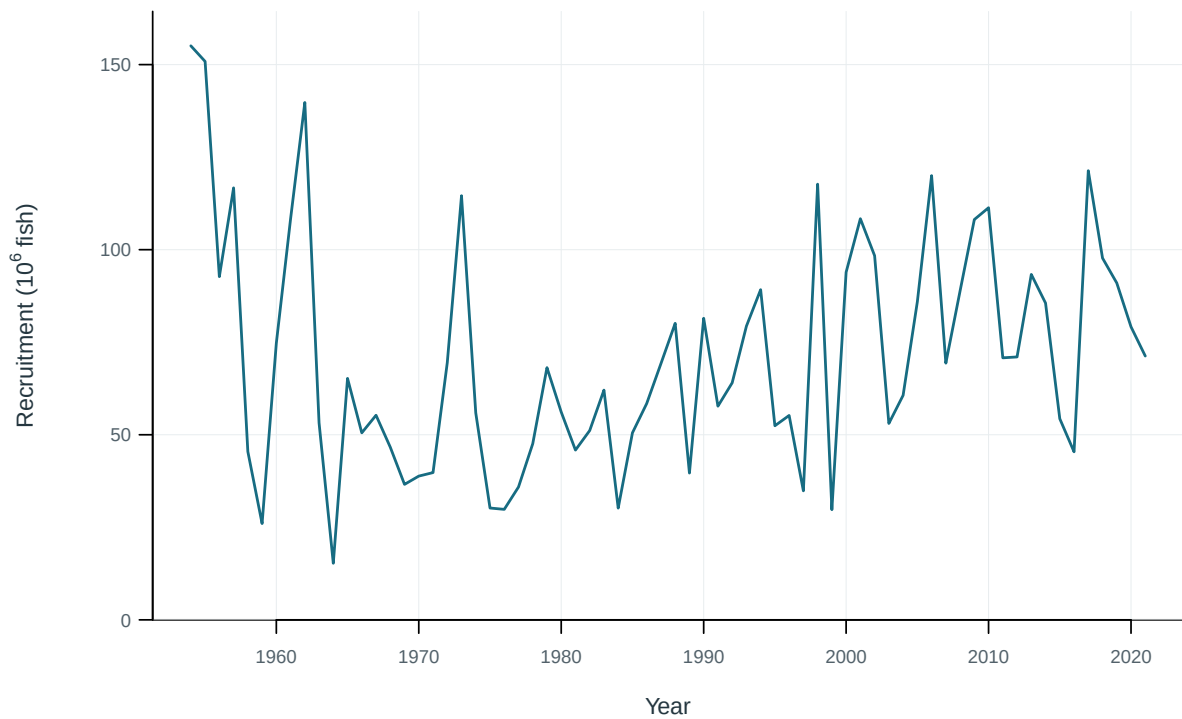


Figure 40: Annual recruitment.

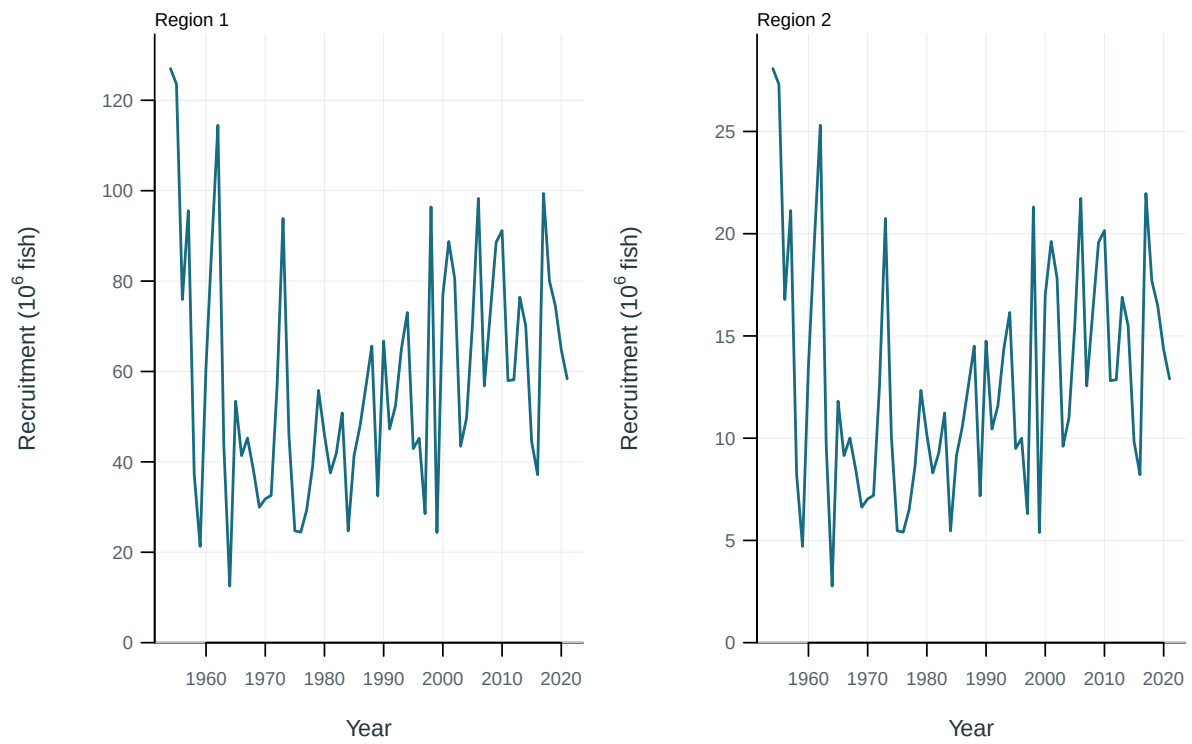


Figure 41: Recruitment by model region.

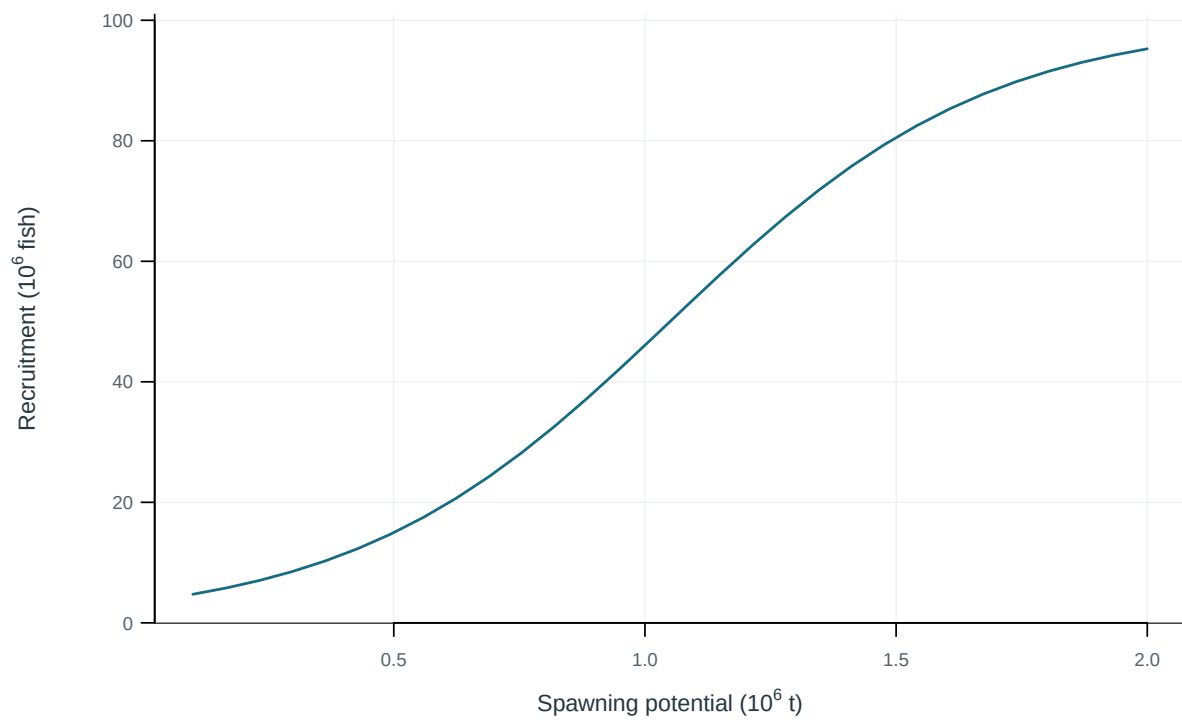


Figure 42: Illustrative output. Stock–recruitment observations and fitted relationship.

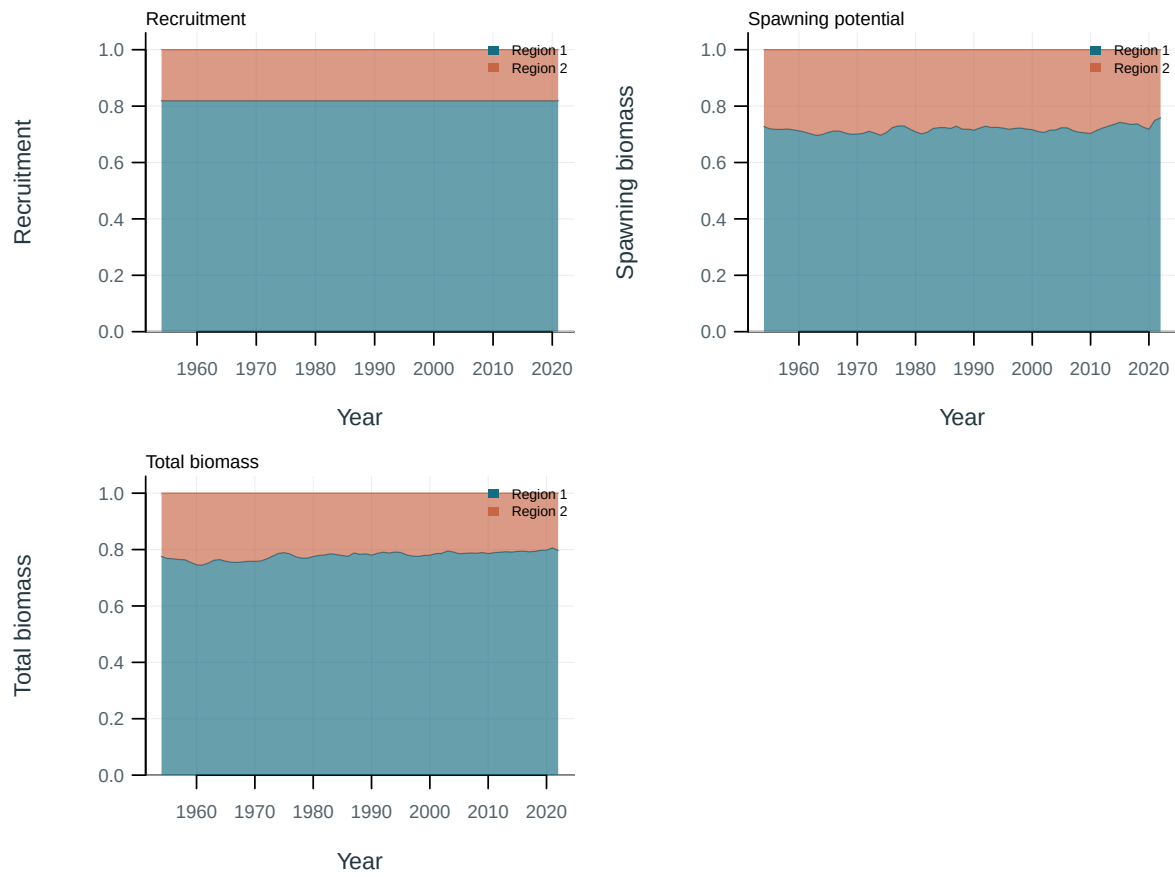


Figure 43: Regional shares of population quantities.

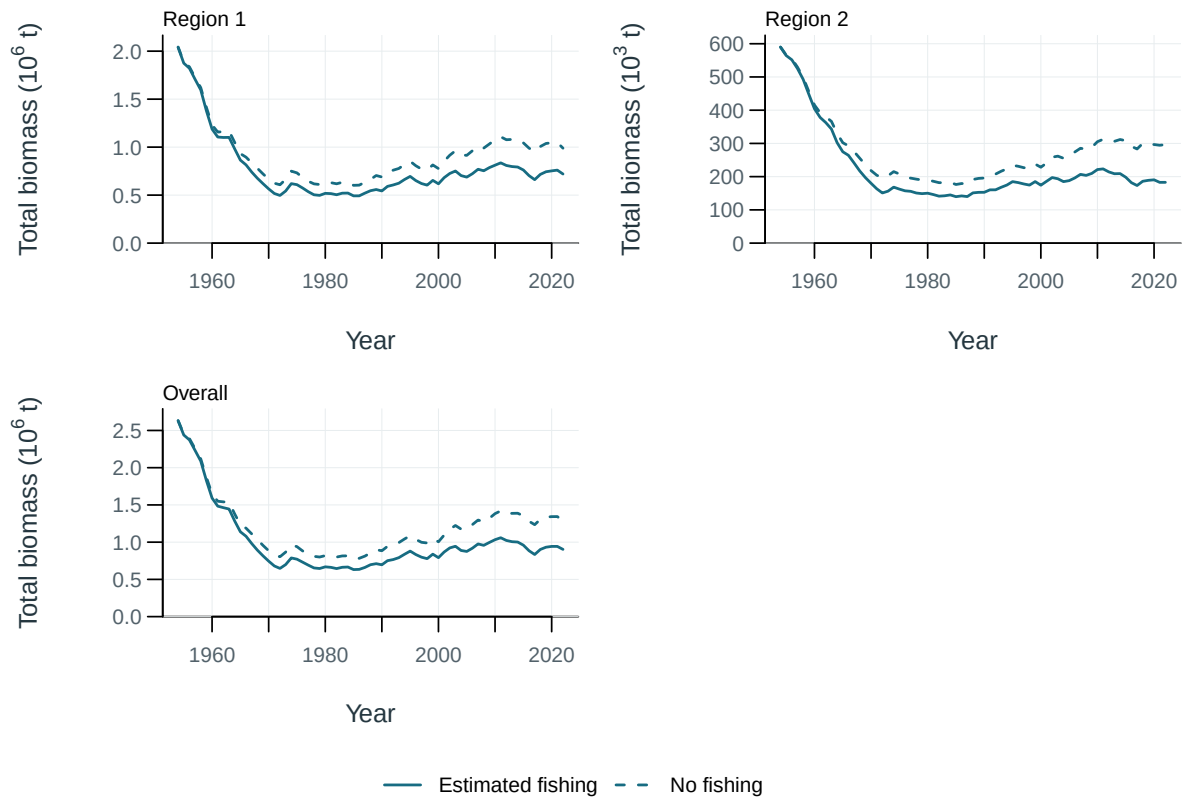


Figure 44: Total biomass under estimated fishing (solid) and the dynamic no-fishing counterfactual (dashed), by region and overall.

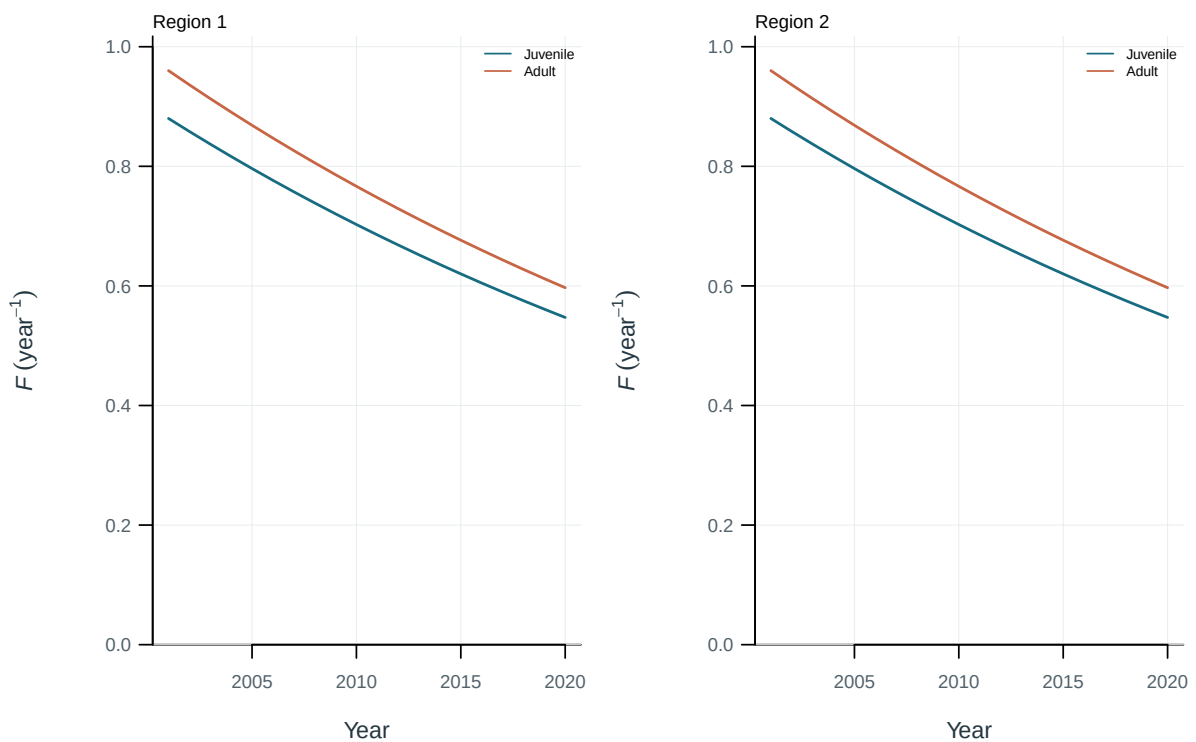


Figure 45: Illustrative output. Fishing mortality by life stage and region.

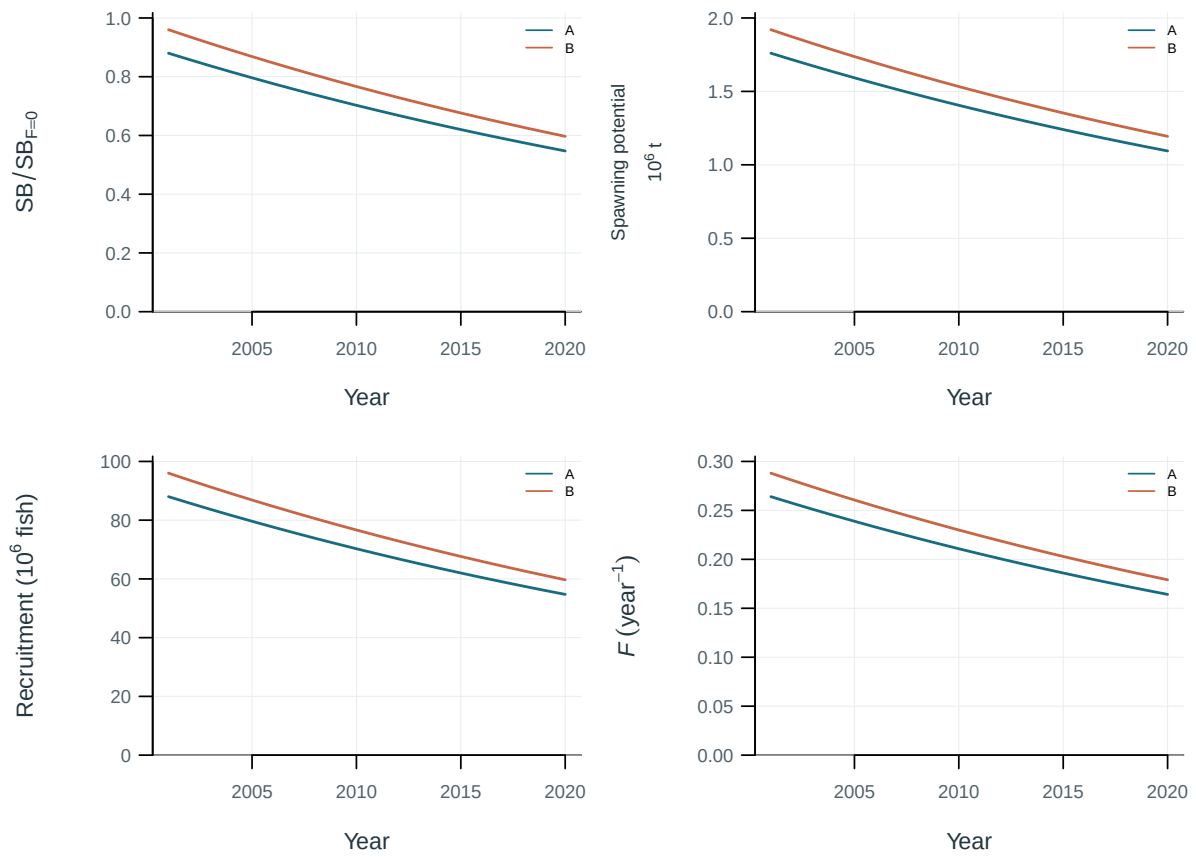


Figure 46: Sensitivity of population trajectories to steepness.

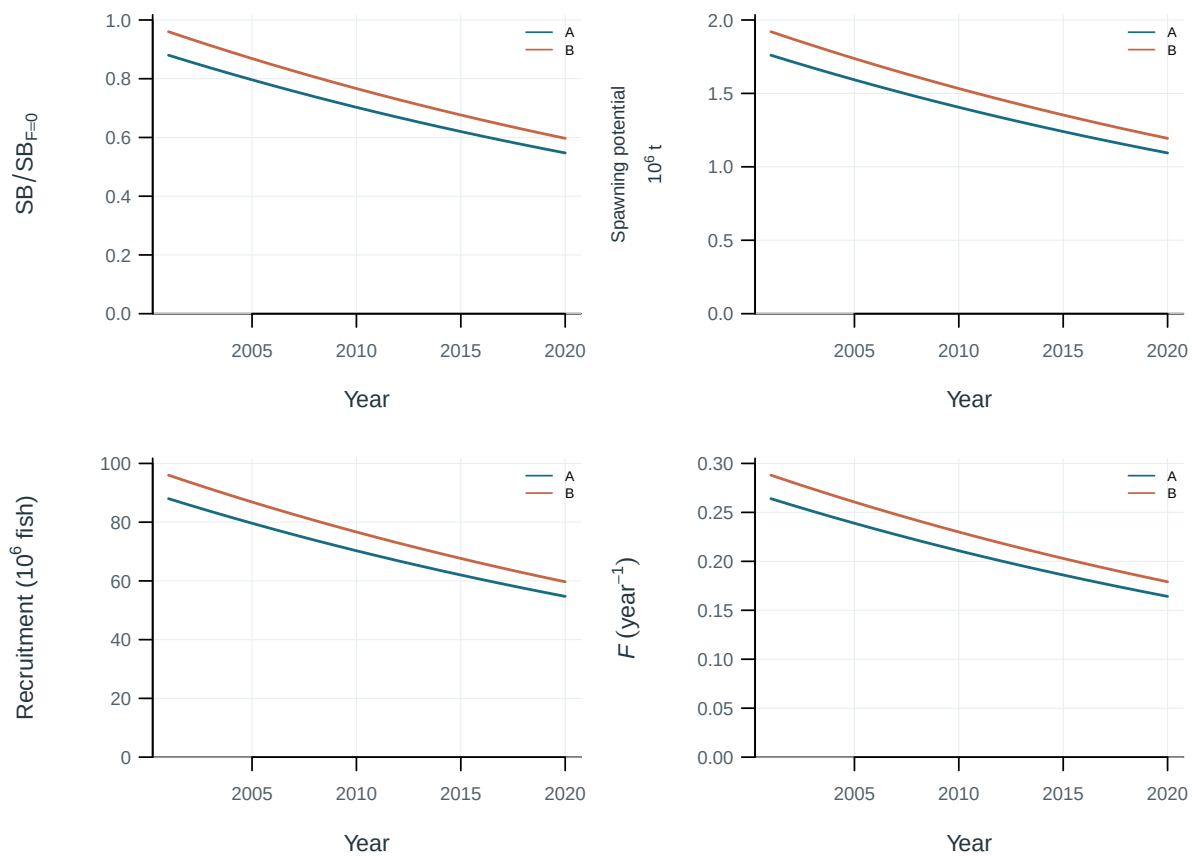


Figure 47: Sensitivity of population trajectories to natural mortality.

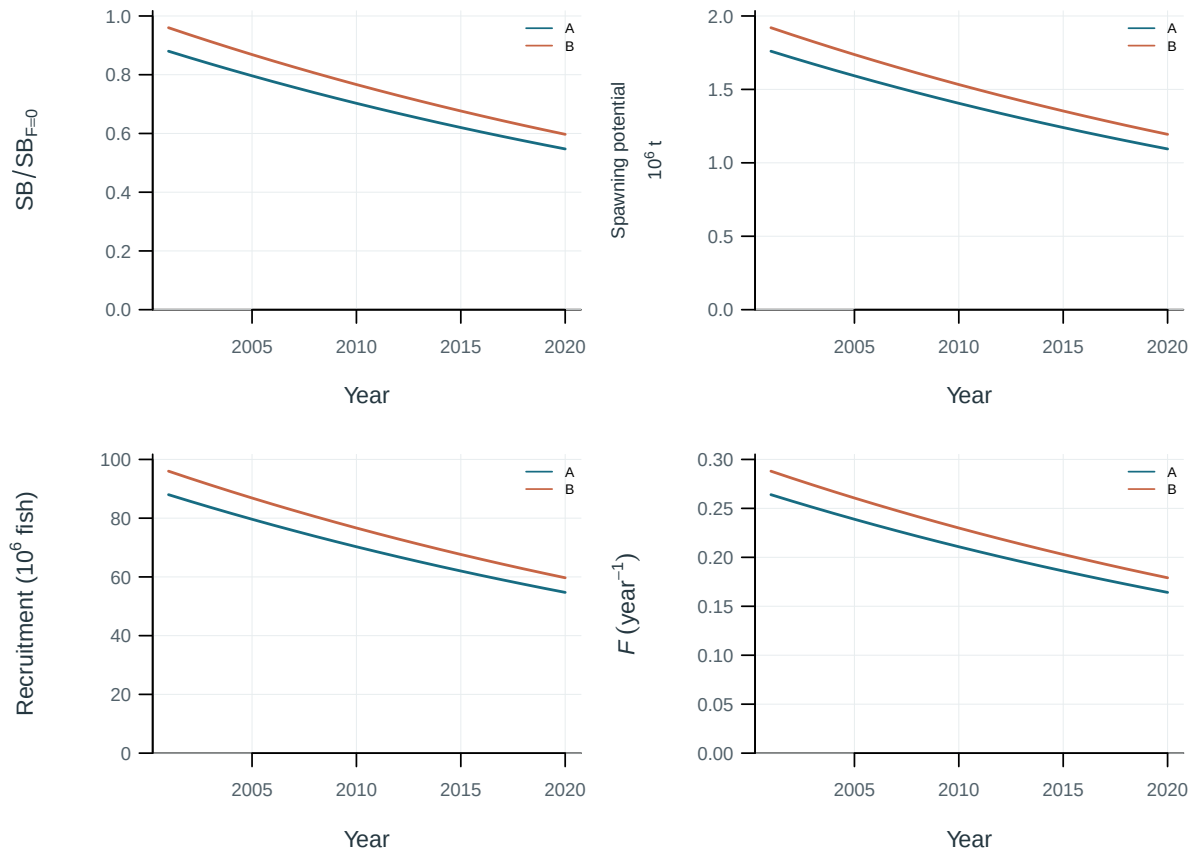


Figure 48: Sensitivity of population trajectories to conditional age-at-length data.

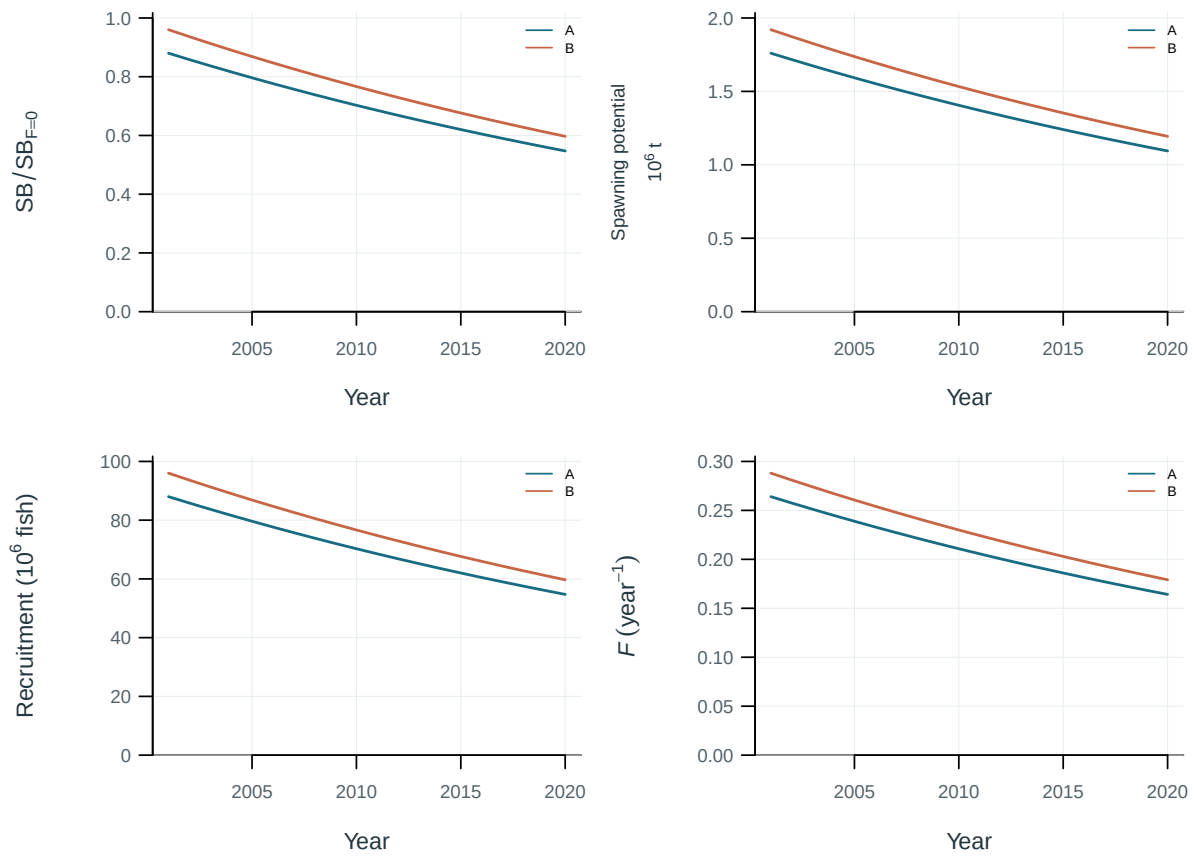


Figure 49: Sensitivity of population trajectories to effort creep.

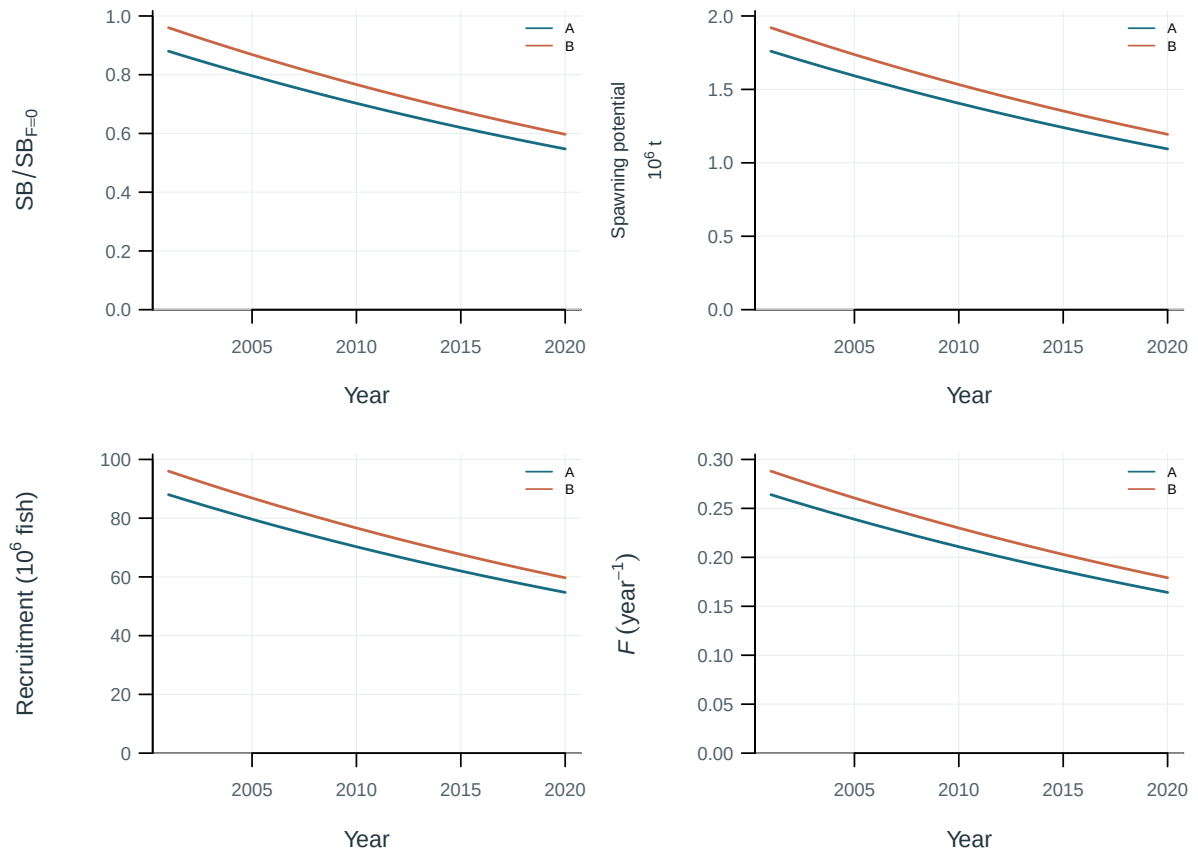


Figure 50: Sensitivity of population trajectories to regional scaling.

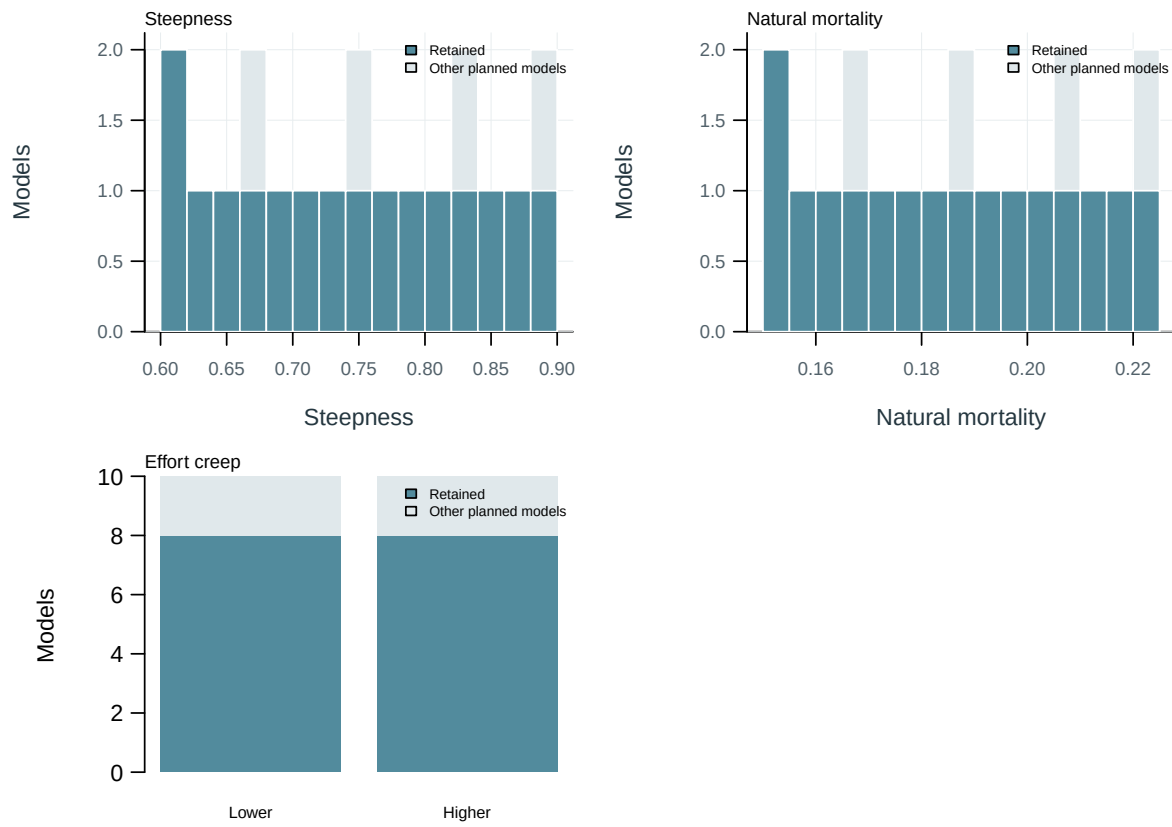


Figure 51: Planned and retained model inputs.

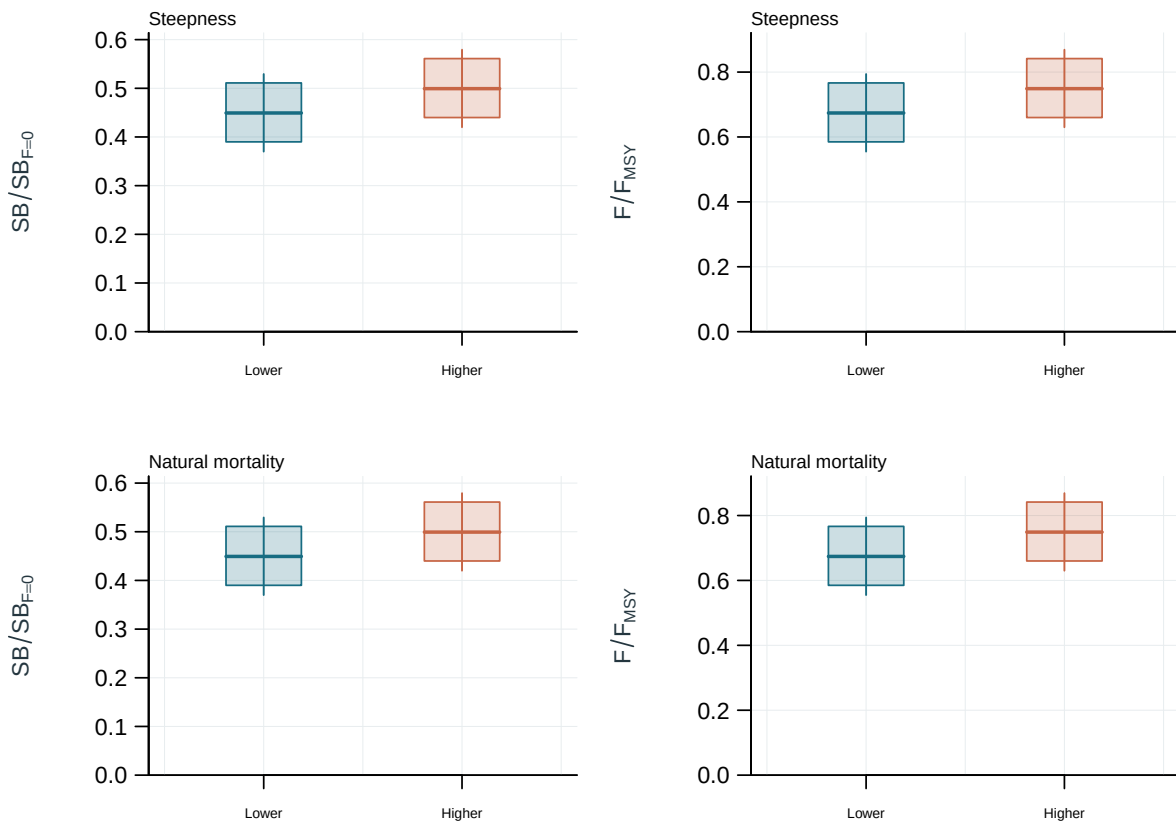


Figure 52: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

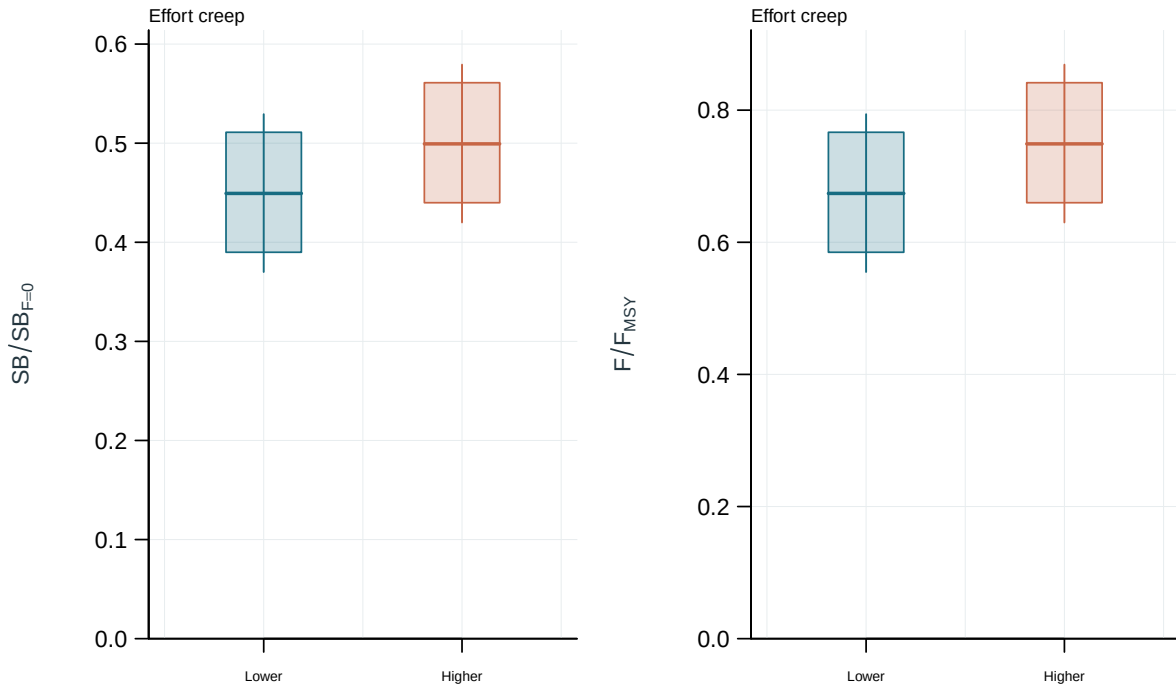


Figure 53: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

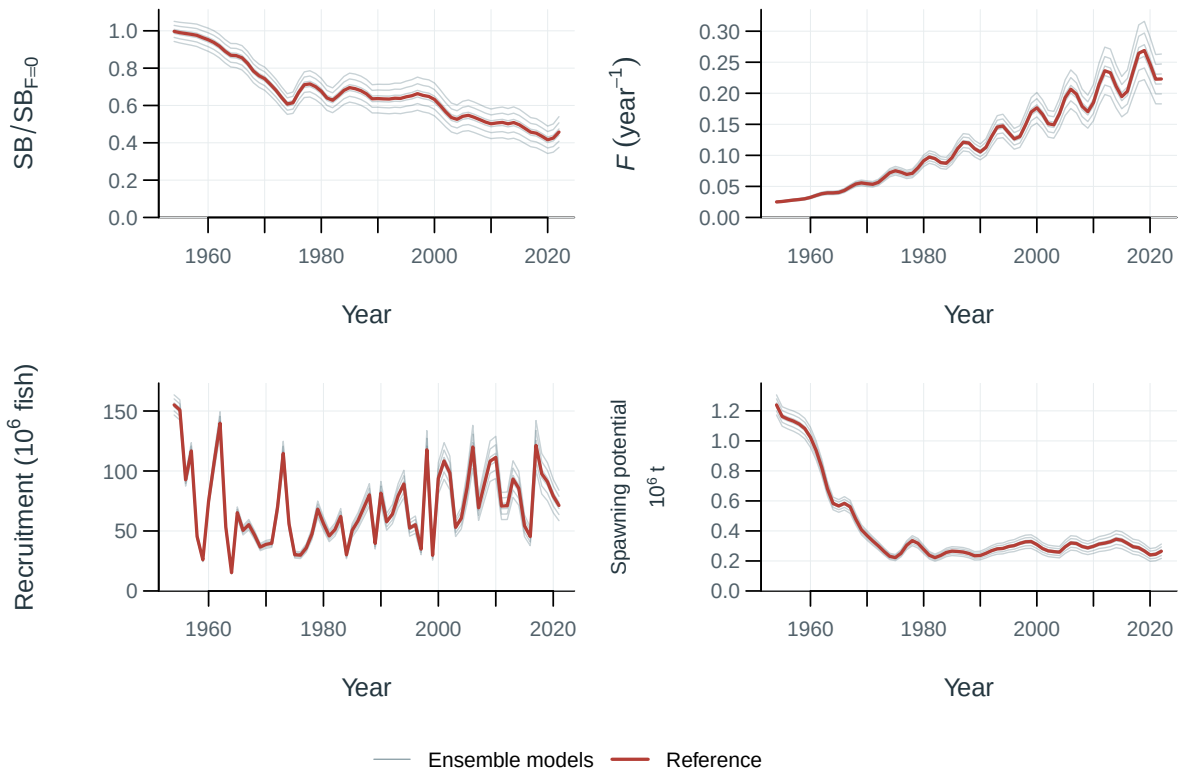


Figure 54: Population trajectories across ensemble models (thin lines) and the reference model (red).

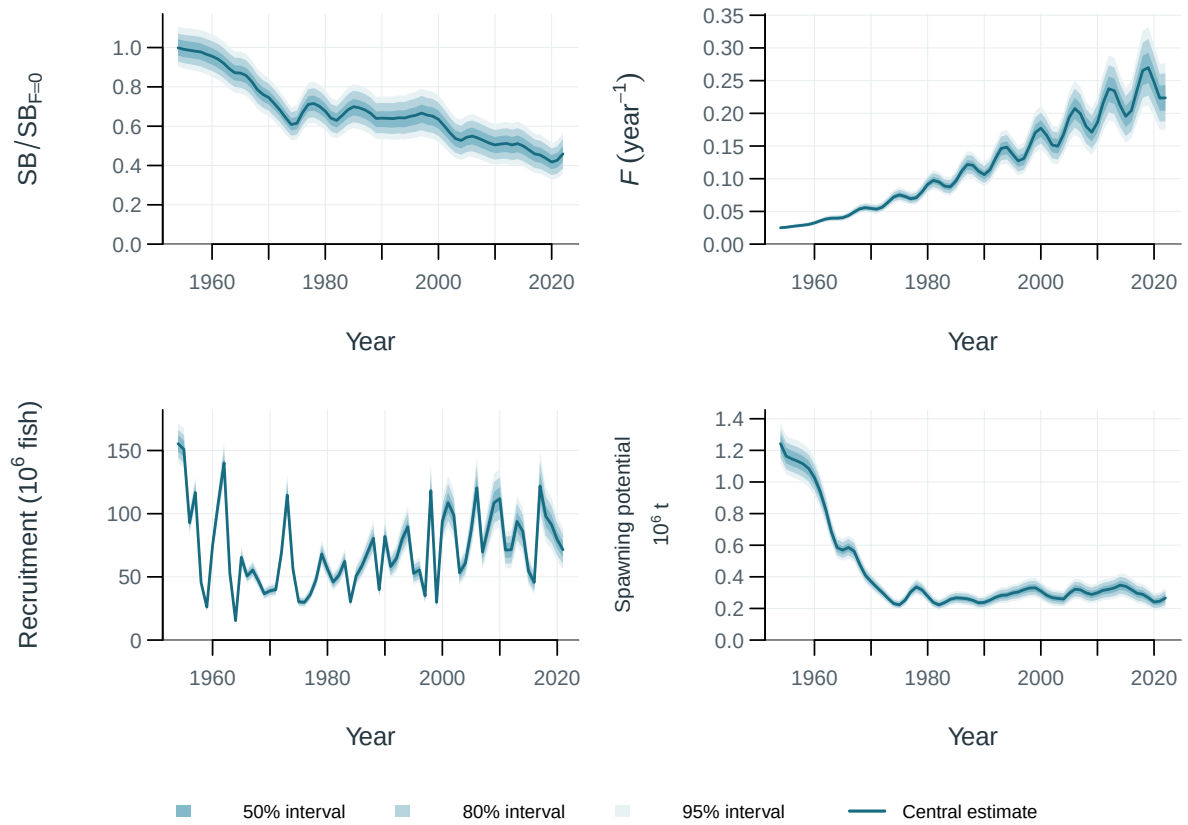


Figure 55: Population trajectories with supplied uncertainty intervals. Central 50, 80, 95% intervals (illustrative model spread and parameter perturbations).

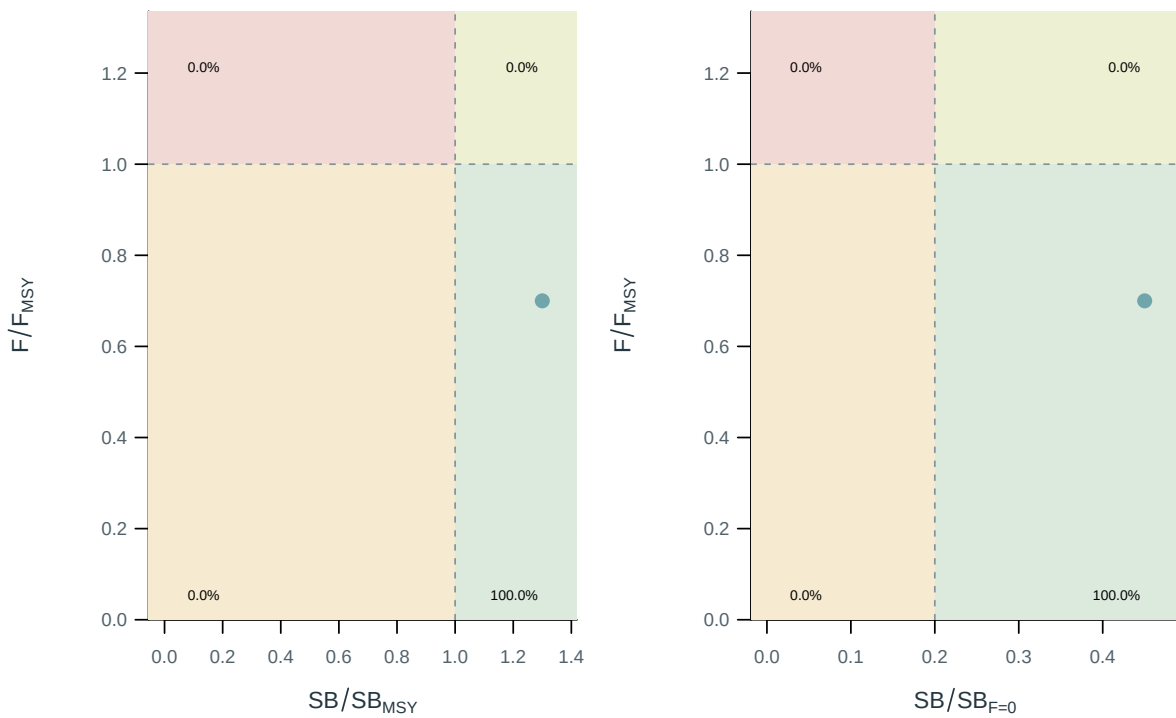


Figure 56: Stock status relative to the supplied spawning and fishing reference points.

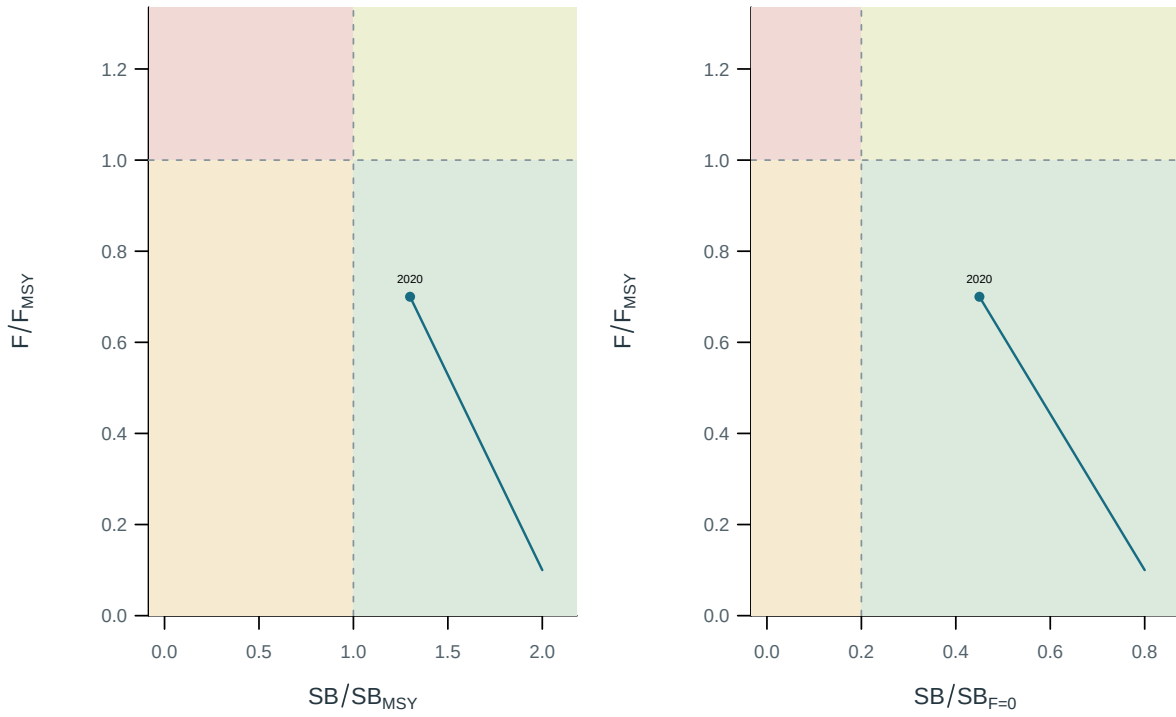


Figure 57: Stock status relative to the supplied spawning and fishing reference points.



Figure 58: Dynamic maximum sustainable yield and catch.

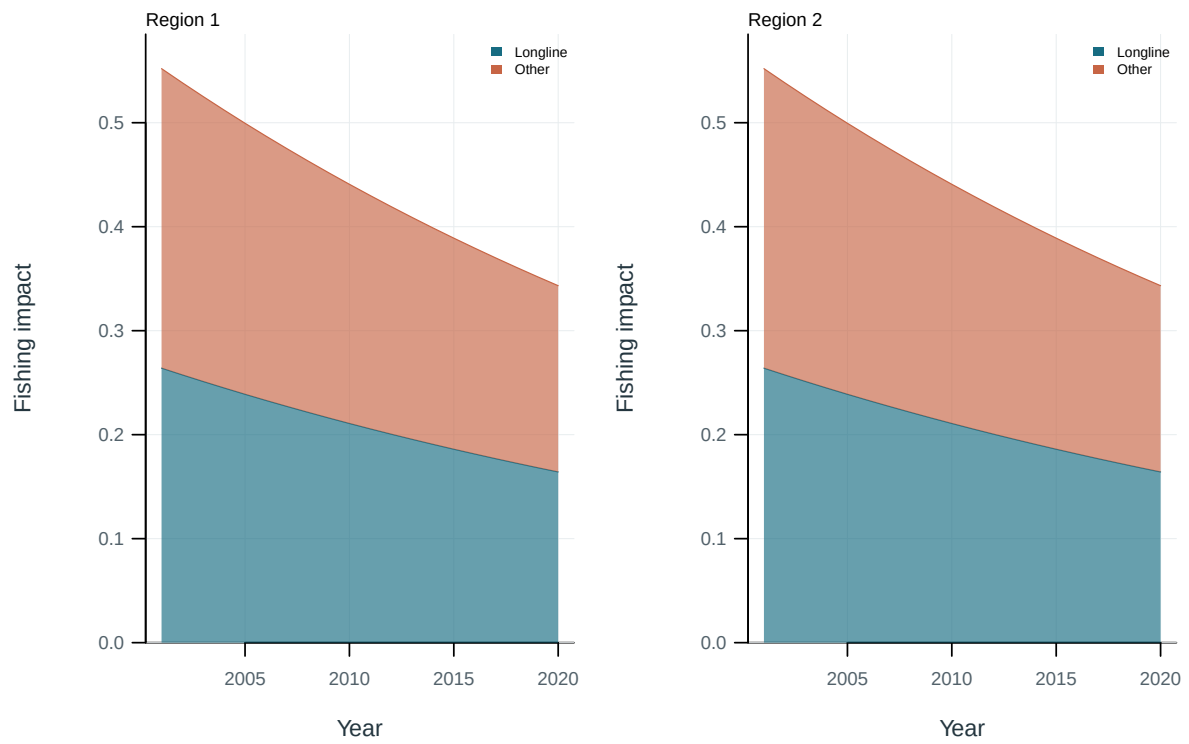


Figure 59: Illustrative output. Contributions of fishing groups to reduced spawning potential.

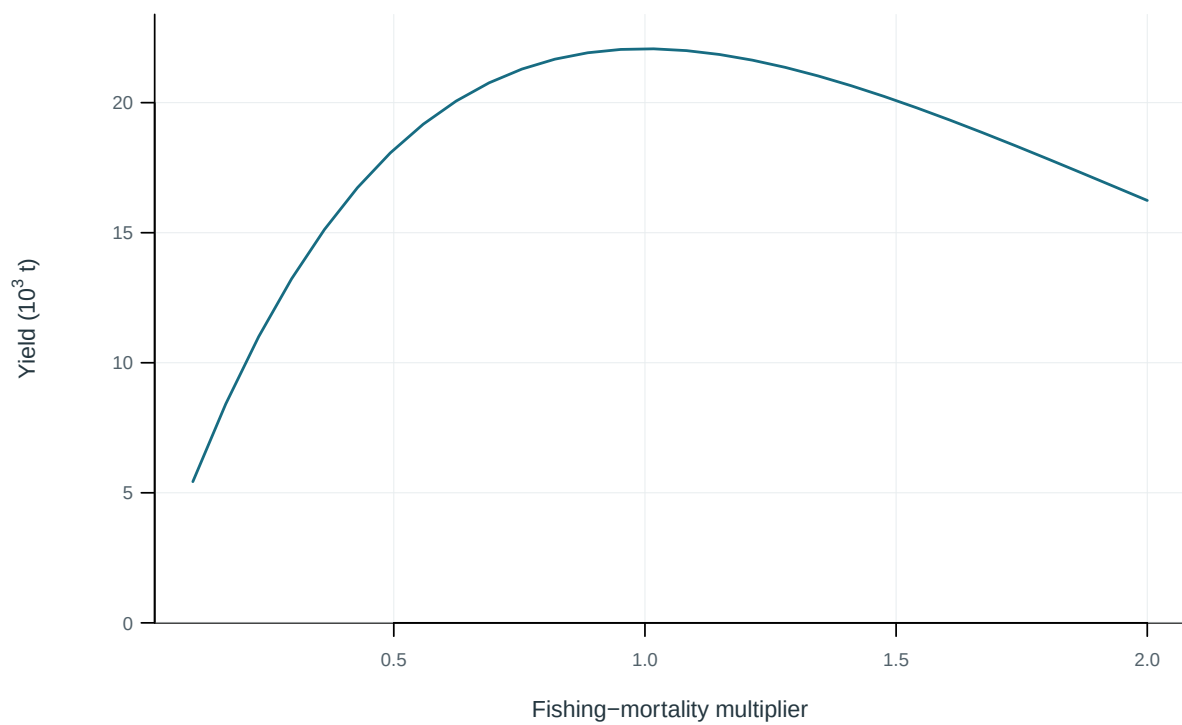


Figure 60: Equilibrium yield across fishing-mortality multipliers.

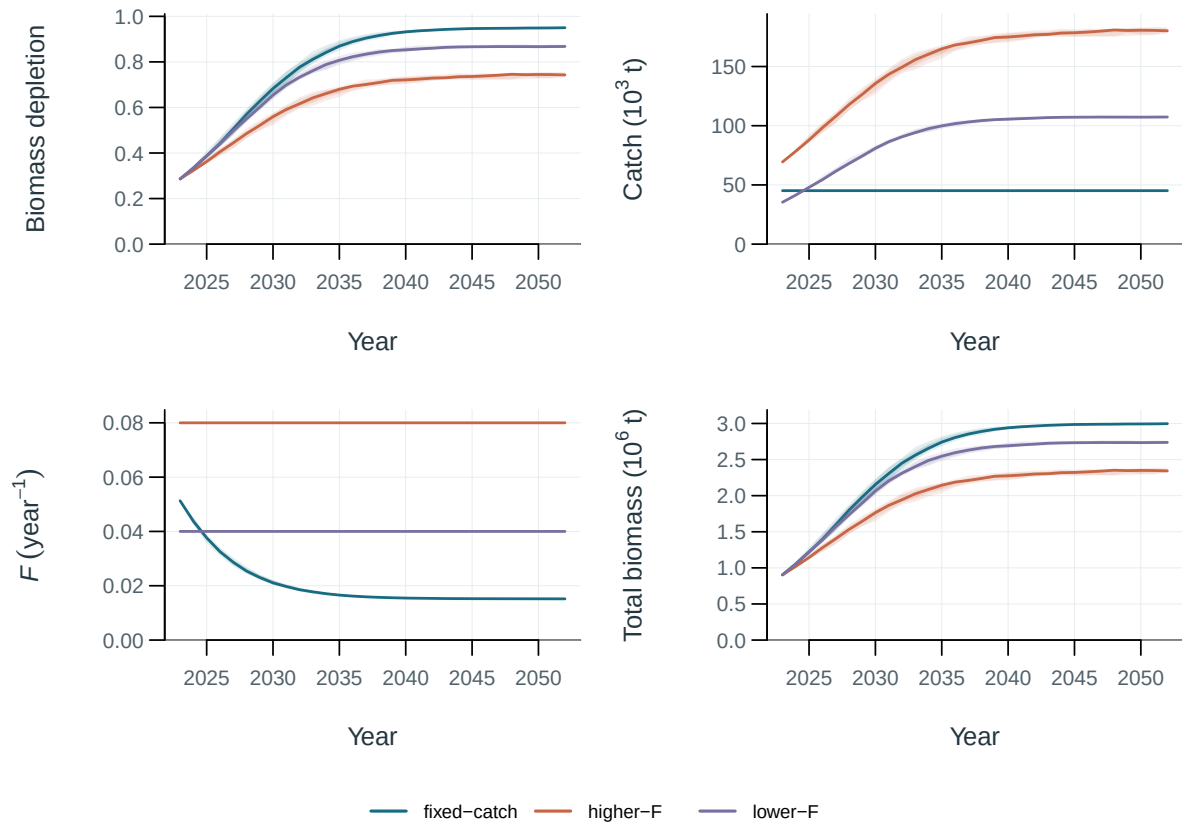


Figure 61: Illustrative surplus-production projections under fixed catch and two F scenarios. Lines are medians; bands span the central 50%, 80% and 95% of production-variability replicates.

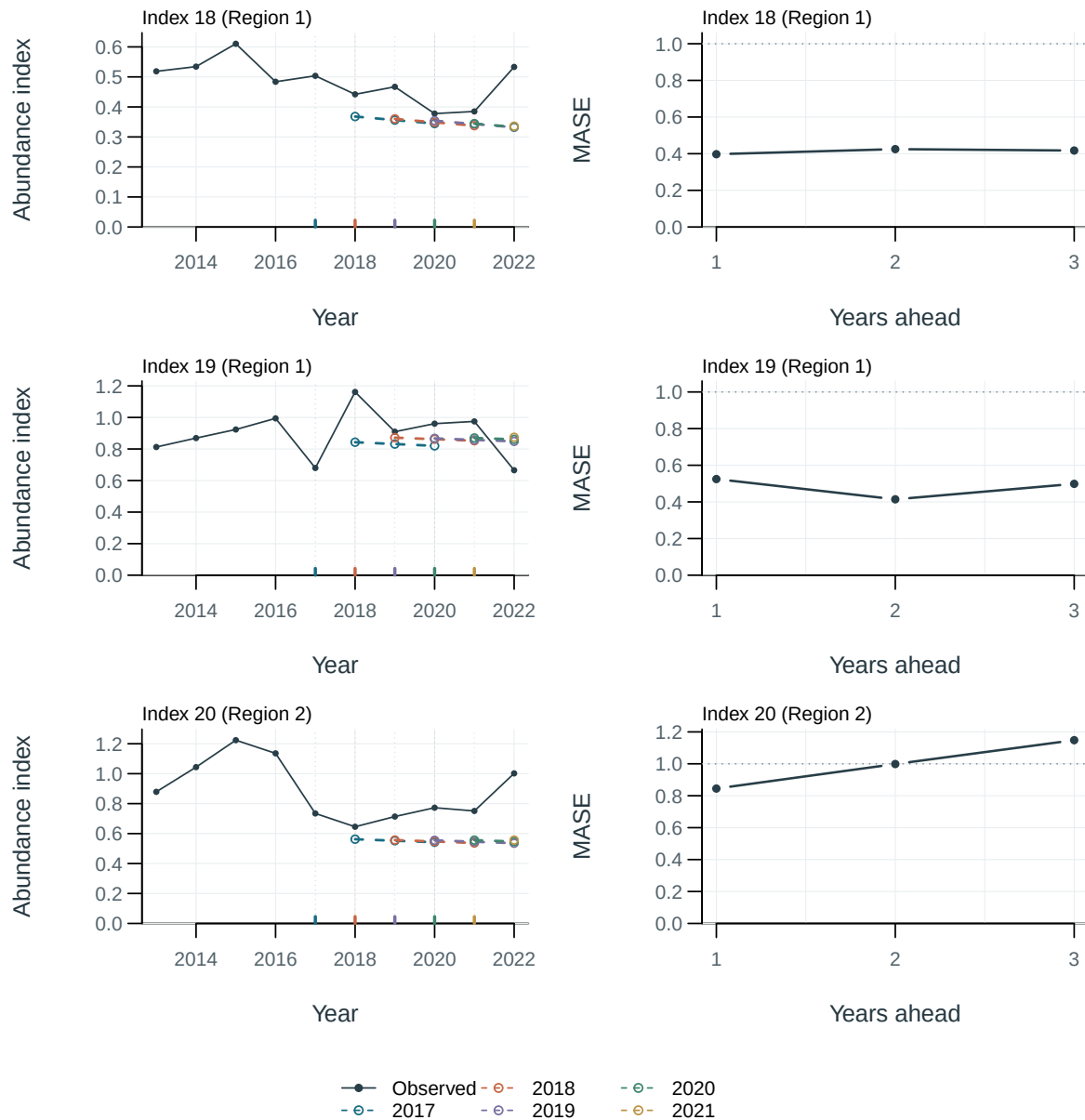


Figure 62: Observations (black) and held-out predictions (coloured dashed lines); vertical marks identify training cutoffs. Right panels show mean absolute scaled error (MASE) by forecast lead. The dotted line marks the training naive-error scale of one; lower values indicate smaller errors.

11 Supplementary outputs

[ALB · Diagnostic model](#)

[ALB · Diagnostic model figures and tables](#)

[Stepwise development](#)

[Stepwise development figures and tables](#)

[Jitter analysis](#)

[Jitter analysis figures and tables](#)

[ALB · Self-test recovery](#)

[ALB · Self-test recovery figures and tables](#)

[ALB · Retrospective analysis](#)

[ALB · Retrospective analysis figures and tables](#)

[ALB · Sensitivities](#)

[ALB · Sensitivities figures and tables](#)

[ALB · Uncertainty ensemble](#)

[ALB · Uncertainty ensemble figures and tables](#)

[Objective profiles · Display example](#)

[Objective profiles · Display example figures and tables](#)

[Additional output formats · Display examples](#)

[Additional output formats · Display examples figures and tables](#)

[Hindcasting](#)

[Hindcasting figures and tables](#)

[Projection scenarios](#)

[Projection scenarios figures and tables](#)

12 Appendix: supplementary methods

13 References